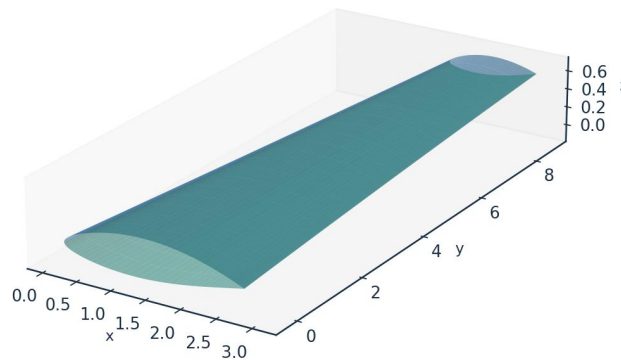


INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

AKOSA SAMUEL ONYEJEKWE

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, $M=0.42$, $Re_{MAC} \approx 6.4 \times 10^6$):

Quantity	Predicted value
Upper-surface transition x_{tr}/c	0.45 (natural / Tollmien–Schlichting)
Lower-surface transition x_{tr}/c	0.58 (natural / Tollmien–Schlichting)
Mean laminar-flow extent	$\approx 52\%$ of chord
Section lift coefficient C_l	0.502
Section profile drag C_d	49.6 counts
Viscous drag reduction vs fully-turbulent	$\approx 43\%$
Climb ($Tu=0.9\%$) transition x_{tr}/c (upper)	0.05 (bypass)

Table 1. Headline predictions.

Validated against three independent, credible published transition datasets with one universal calibration set, the solver reproduces the transition-onset Reynolds number $Re_{\theta t}$ against the Rolls-Royce hot-wire measurements of ERCOFTAC case 020, the Schubauer-Skramstad experiment, and two independent swept-wing experiments. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f , momentum thickness θ , shape factor H , intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e , with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at $M = 0.42$ against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below $M = 0.2$.
- Laminar boundary layer: Thwaites' integral method advances θ , H and Re_θ from the stagnation point along each surface, with the fluid properties evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction and momentum thickness.

- Unified transition kernel (the novel core): at every station the effective transition Reynolds number is the minimum across the four mechanisms, each with a calibration weight.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ .
- Turbulent boundary layer: Head's entrainment method with the Ludwig–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns profile drag; a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the $e^{\Lambda N}$ integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at $Re_{\theta} = 200$ against the accepted 200.5, a shape factor of 2.59129 and a wall shear parameter $f''(0) = 0.46960$. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment and therefore spans measured bubbles from about 40 momentum thicknesses at 2.1 % free-stream turbulence to about 180 at 0.03 %.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethé & Chow)

$$\frac{V_{t_i}}{U_{\infty}} = \cos(\theta_i - \alpha) + \sum_{j=1}^N C_{t,ij} \gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left(\frac{V}{U_{\infty}} \right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_{\infty}^2}{1 + \beta} \right) \frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_{\infty}^2}$$

4.2 Laminar boundary layer (Thwaites)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6} \int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu} \frac{dU_e}{dx}, Re_\theta = \frac{U_e \theta}{\nu}$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2 + H) \frac{\theta}{U_e} \frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu ; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_{crit} ; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset uses the C1 criterion. The kernel takes the minimum effective onset across all four:

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N , envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^x \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \geq N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U - c)(D^2 - \alpha^2) - U'']\hat{v} = \frac{1}{i\alpha Re_\theta} (D^2 - \alpha^2)^2 \hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma = -\alpha_i \theta = \frac{\omega_i}{c_g}, c_g = \frac{\partial \omega_r}{\partial \alpha_r}$$

(E11) Cross-flow criterion (swept wing, C1 on $Re_{\theta 2}$)

$$Re_{\theta 2} = k_{cf} Re_\theta \sin \Lambda \cos \Lambda \geq C_1$$

(E12) Separation bubble: dead-air momentum march

$$\frac{d\theta}{dx} = -(2 + H_s) \frac{\theta}{U_e} \frac{dU_e}{dx}, C_f = 0, H = H_s$$

(E12b) Separation bubble: reattachment length

$$x_r - x_s = \frac{N_{crit} \theta_s}{\sigma_{sep}}, \sigma_{sep} = 0.048$$

(E13) Unified minimum-onset transition kernel

$$Re_{\theta t}^* = \min(a_{TS} Re_{\theta t}^{TS}, a_{BP} Re_{\theta t}^{BP}, a_{SEP} Re_{\theta t}^{SEP}, a_{CF} Re_{\theta t}^{CF})$$

(E14) Transition trigger

$$Re_{\theta}(x_t) \geq Re_{\theta t}^*$$

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x) = 1 - \exp[-0.412 \xi^2], \xi = \frac{x - x_t}{\lambda_{tr}}$$

(E16) Transition-length scale

$$\lambda_{tr} \sim \frac{\nu}{U_e} C_{len} Re_{\theta t}^{0.8}$$

(E17) Intermittency-weighted property blend

$$\phi = (1 - \gamma) \phi_{lam} + \gamma \phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e} \frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwig-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_{\theta}^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2 \frac{\theta_{TE}}{c} \left(\frac{U_{e,TE}}{U_{\infty}} \right)^{(H_{TE}+5)/2}$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r \frac{\gamma - 1}{2} M_e^2 \left[1 - \left(\frac{u}{U_e} \right)^2 \right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e \left(1 + r \frac{\gamma - 1}{2} M_e^2 \right), r \approx Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032 M_e^2 + 0.58 \left(\frac{T_w}{T_e} - 1 \right), \frac{\nu_{ref}}{\nu_e} = \left(\frac{T_{ref}}{T_e} \right)^{1+\omega}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_{\infty} U_{\infty} \bar{c}}{\mu_{\infty}} = \frac{U_{\infty} \bar{c}}{\nu_{\infty}}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e ^N codes	RANS γ-Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e ^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design loops	Partly	No	No	Yes

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally by taking the minimum effective onset Reynolds number across four co-resident mechanisms, each modulated by a calibration weight, and feeds a single intermittency closure. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. The separation and cross-flow branches are carried and calibrated but are not selected at any condition reported here — the novelty claim.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-

Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mac_h	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (FL360, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (FL100, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input).

Flight-condition comparison: cruise vs climb (flow_conditions.csv)

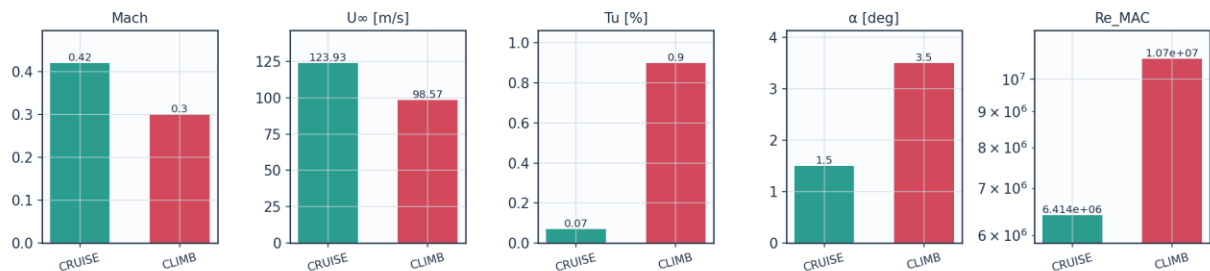


Fig. 8a. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K

Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m ³

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) bypass(AGS@Tu) separation crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann Cf
Separation criterion	Thwaites $\lambda \leq -0.09$ / $H > 2.6$ turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset weight	unity; validated on Schubauer-Skramstad and NLF(1)-0416 upper surface
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e ^N amplification factor	Mack N=-8.43-2.4 ln(Tu), clamped to 0.0008 ≤ Tu ≤ 0.0298; 8.68 for Tu < 0.08%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_TS_max	0.1	upper Tu for the attached-flow TS branch	complement of the bypass gate
C_len	6.5	Narasimha transition-length scale	Dhawan-Narasimha (1958)
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al. C1 criterion; does not transfer between facilities
CF_ratio	0.47	theta2/theta surrogate for crossflow	set on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	0.048	shear-layer amplification rate in a bubble	set on T3C4 and NLF(1)-0416 bubble measurements
n_freq	40	frequencies carried by the e ^N integration	discretisation parameter; onset moves <0.2% over

Table 7. Calibration constants and their sources (input).

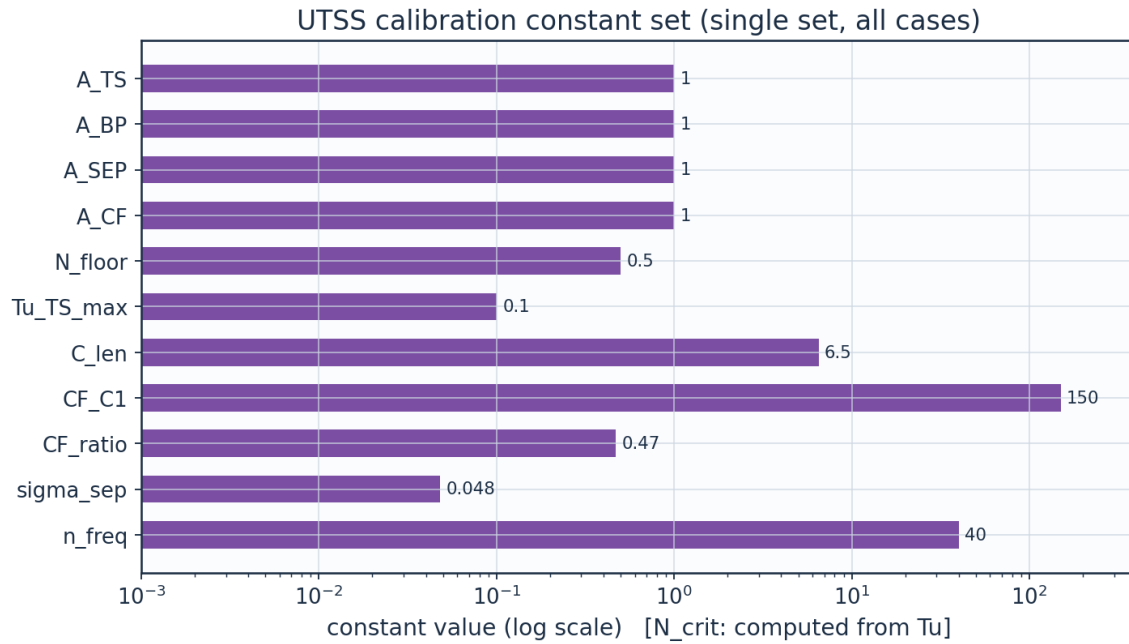


Fig. 8b. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

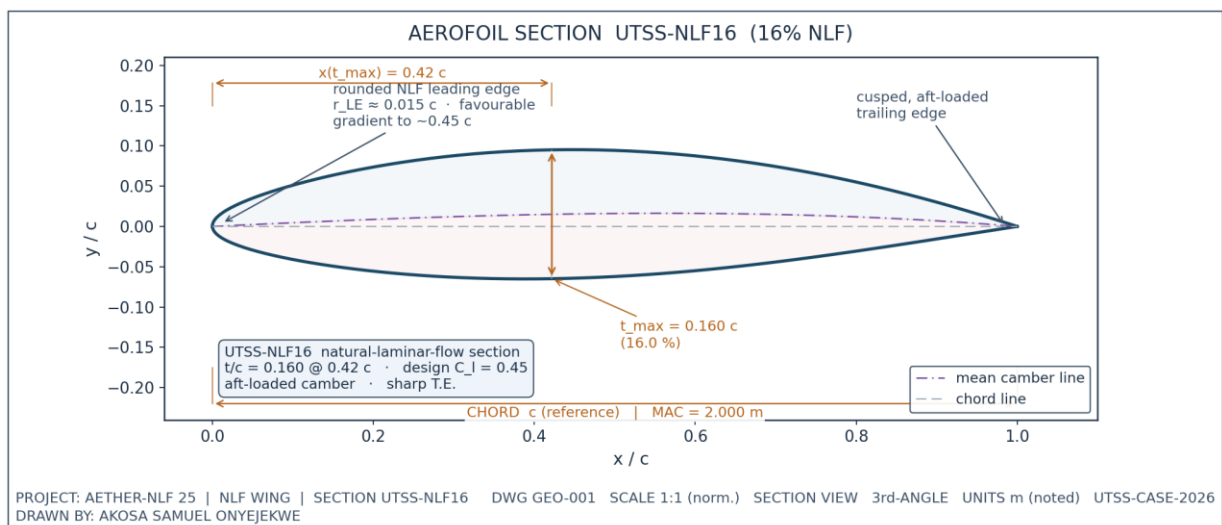


Fig. 1. UTSS-NLF16 aerofoil section (dimensioned).

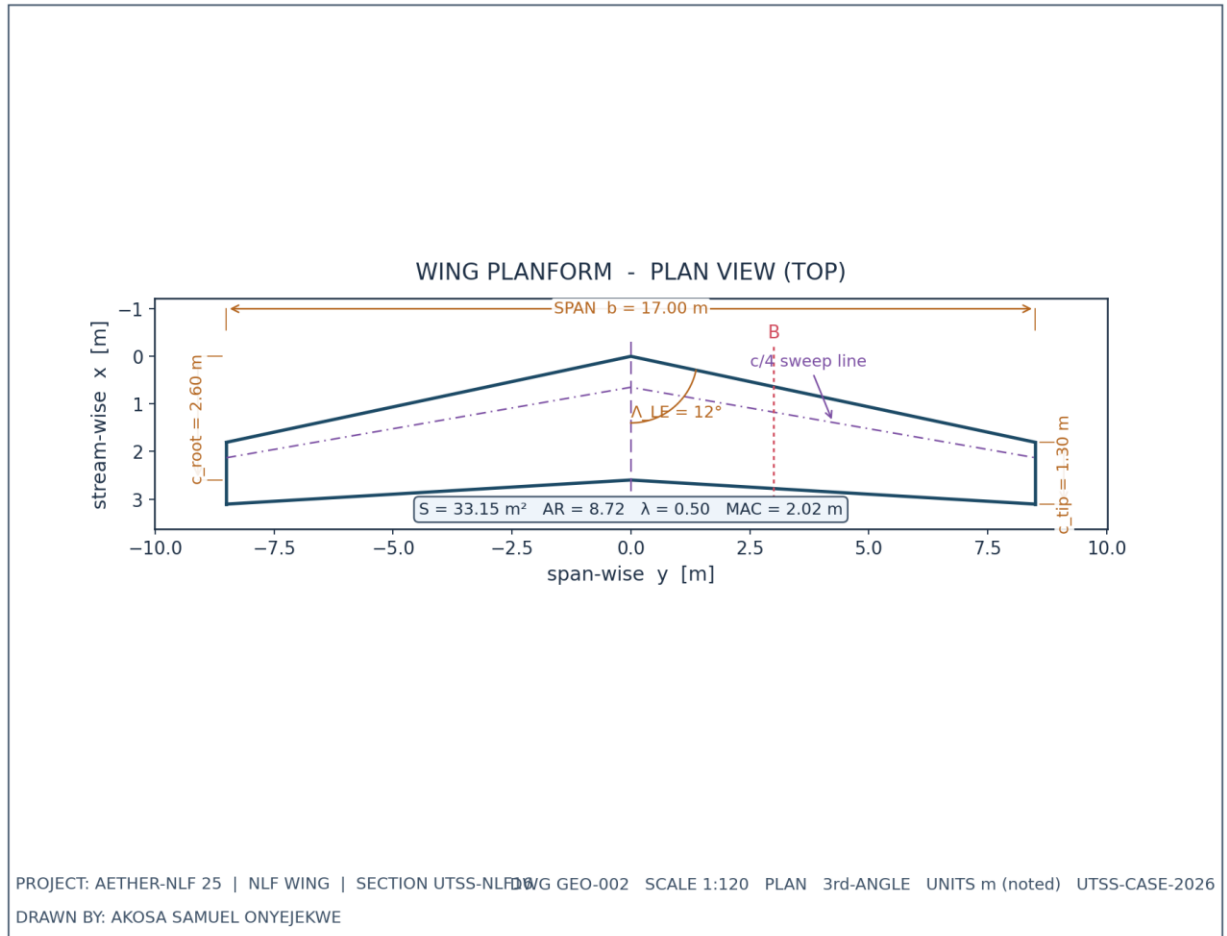


Fig. 2. Wing planform — plan (top) view, dimensioned.

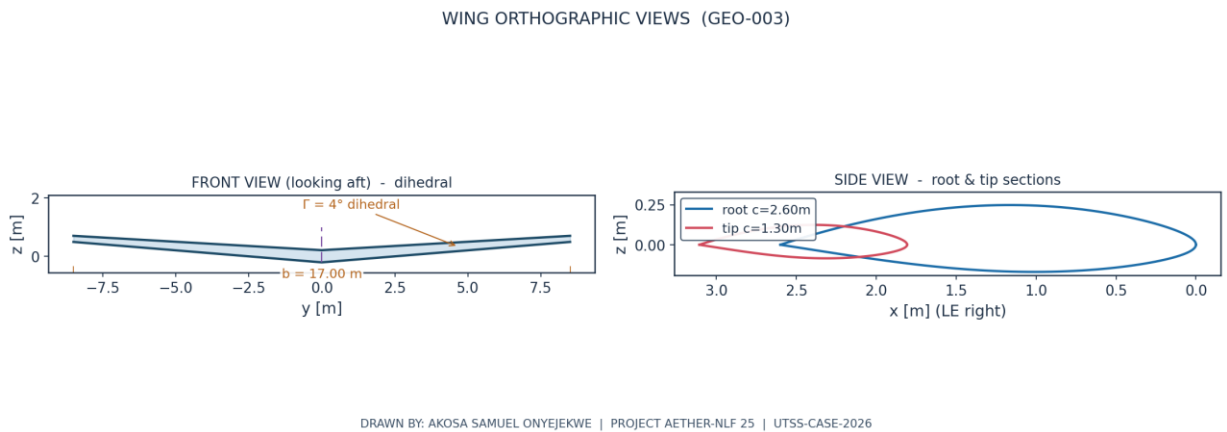
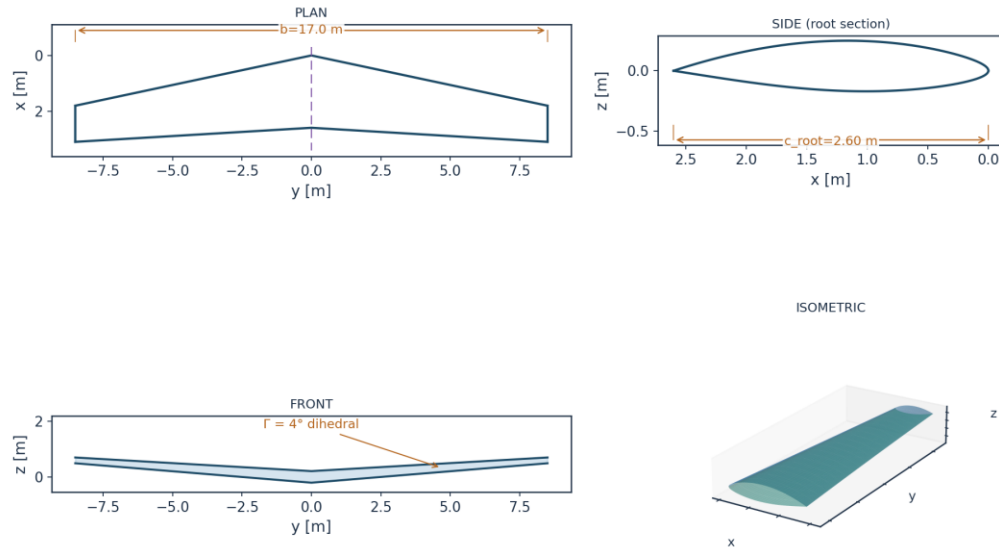


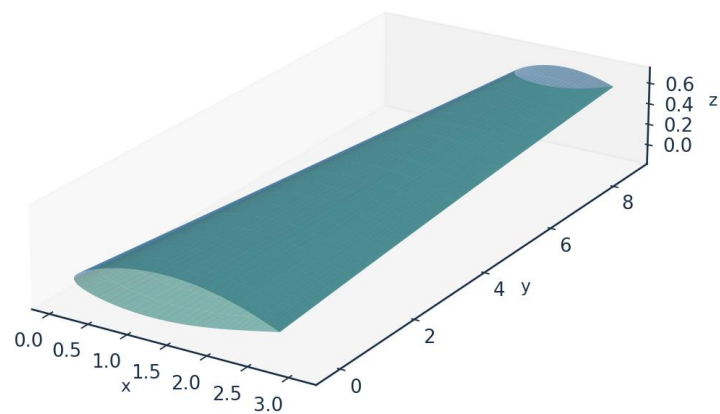
Fig. 3. Front and side orthographic views.



All dimensions in metres unless noted | Scale 1:120 | UTSS-CASE-2026 | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 4. Full third-angle orthographic projection sheet.

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown | DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Fig. 5. Isometric view of the wing.

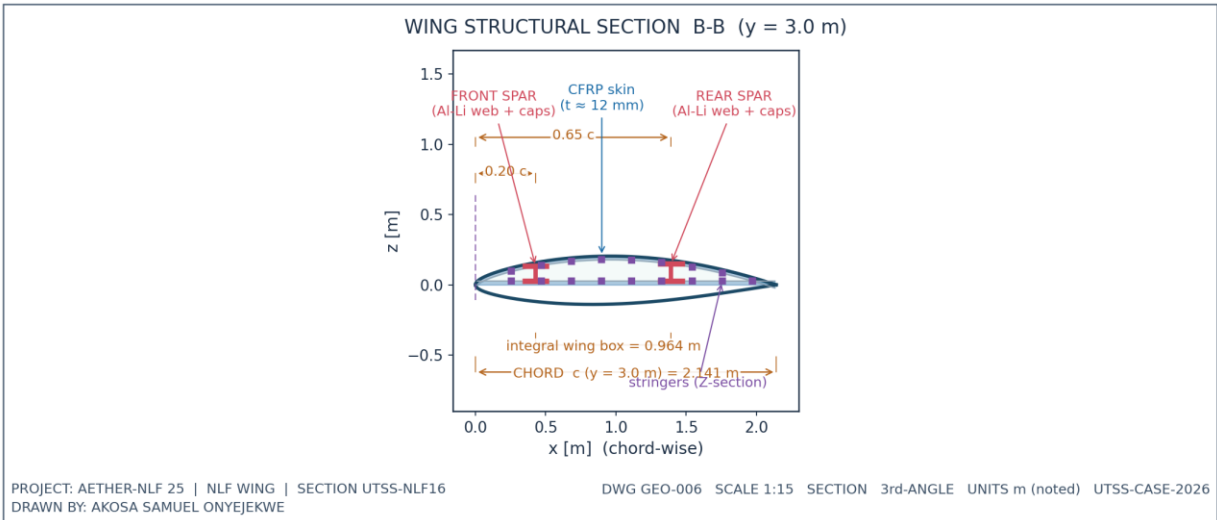


Fig. 6. Structural sectional view B-B.

7.1 Geometry data (from CSV)

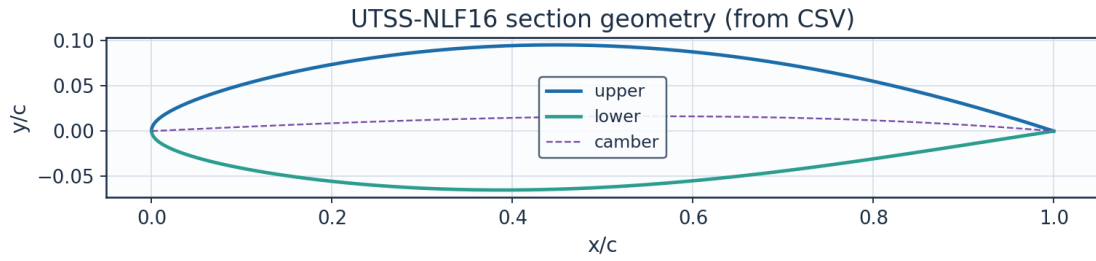


Fig. 7. Section geometry plotted from airfoil_UTSS-NLF16.csv.

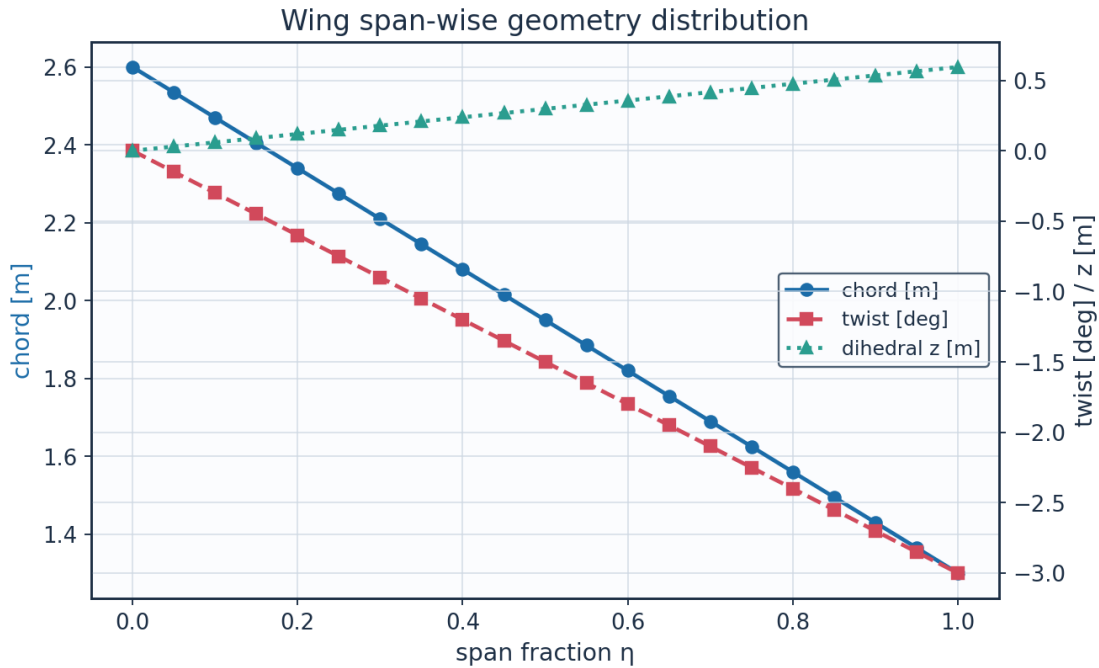


Fig. 8. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m	Re_local
0.0	0.0	2.6	0.0	2.6	-0.0	0.0	8246203.012 995781
0.05	0.425	2.535	0.090336538 7097594	2.625336538 70976	-0.15	0.029718895 0759919	8040047.937 6708865
0.1	0.850000000 0000001	2.47	0.180673077 4195188	2.650673077 419519	-0.3	0.059437790 1519838	7833892.862 345993
0.15	1.275	2.405	0.271009616 1292782	2.676009616 1292783	-0.45	0.089156685 2279757	7627737.787 021098
0.2	1.700000000 0000002	2.34	0.361346154 8390376	2.701346154 8390376	- 0.600000000 0000001	0.118875580 3039677	7421582.711 696202
0.25	2.125	2.275	0.451682693 548797	2.726682693 548797	-0.75	0.148594475 3799596	7215427.636 371308
0.3	2.550000000 0000003	2.21	0.542019232 2585565	2.752019232 2585566	- 0.900000000 0000001	0.178313370 4559515	7009272.561 046414

0.35	2.975	2.145	0.632355770 9683159	2.777355770 968316	-1.05	0.208032265 5319434	6803117.485 721519
0.4	3.400000000 0000004	2.08	0.722692309 6780753	2.802692309 678076	- 1.200000000 0000002	0.237751160 6079354	6596962.410 396625
0.45	3.825	2.015	0.813028848 3878346	2.828028848 387835	-1.35	0.267470055 6839273	6390807.335 07173
0.5	4.25	1.95	0.903365387 097594	2.853365387 097594	-1.5	0.297188950 7599192	6184652.259 7468365
0.55	4.675000000 000001	1.885	0.993701925 8073536	2.878701925 807354	-1.65	0.326907845 8359112	5978497.184 421942
0.600000000 0000001	5.100000000 0000005	1.82	1.084038464 517113	2.904038464 5171127	- 1.800000000 0000005	0.356626740 9119031	5772342.109 097046
0.65	5.525	1.755	1.174375003 2268725	2.929375003 2268724	-1.95	0.386345635 987895	5566187.033 772152
0.700000000 0000001	5.95	1.69	1.264711541 9366315	2.954711541 936632	-2.1	0.416064531 0638869	5360031.958 447257
0.75	6.375	1.625	1.355048080 6463912	2.980048080 646391	-2.25	0.445783426 1398788	5153876.883 122363
0.8	6.800000000 000001	1.56	1.445384619 3561506	3.005384619 35615	- 2.400000000 0000004	0.475502321 2158709	4947721.807 797468
0.850000000 0000001	7.225	1.495	1.535721158 06591	3.030721158 06591	- 2.550000000 0000003	0.505221216 2918628	4741566.732 472573
0.9	7.65	1.43	1.626057696 7756692	3.056057696 775669	-2.7	0.534940111 3678547	4535411.657 14768
0.95	8.075000000 000001	1.365	1.716394235 485429	3.081394235 485429	-2.85	0.564659006 4438467	4329256.581 822785
1.0	8.5	1.3	1.806730774 195188	3.106730774 195188	-3.0	0.594377901 5198385	4123101.506 49789

Table 8. Wing planform stations (wing_planform.csv).

Lofted wing sections (from wing_sections_3d.csv)

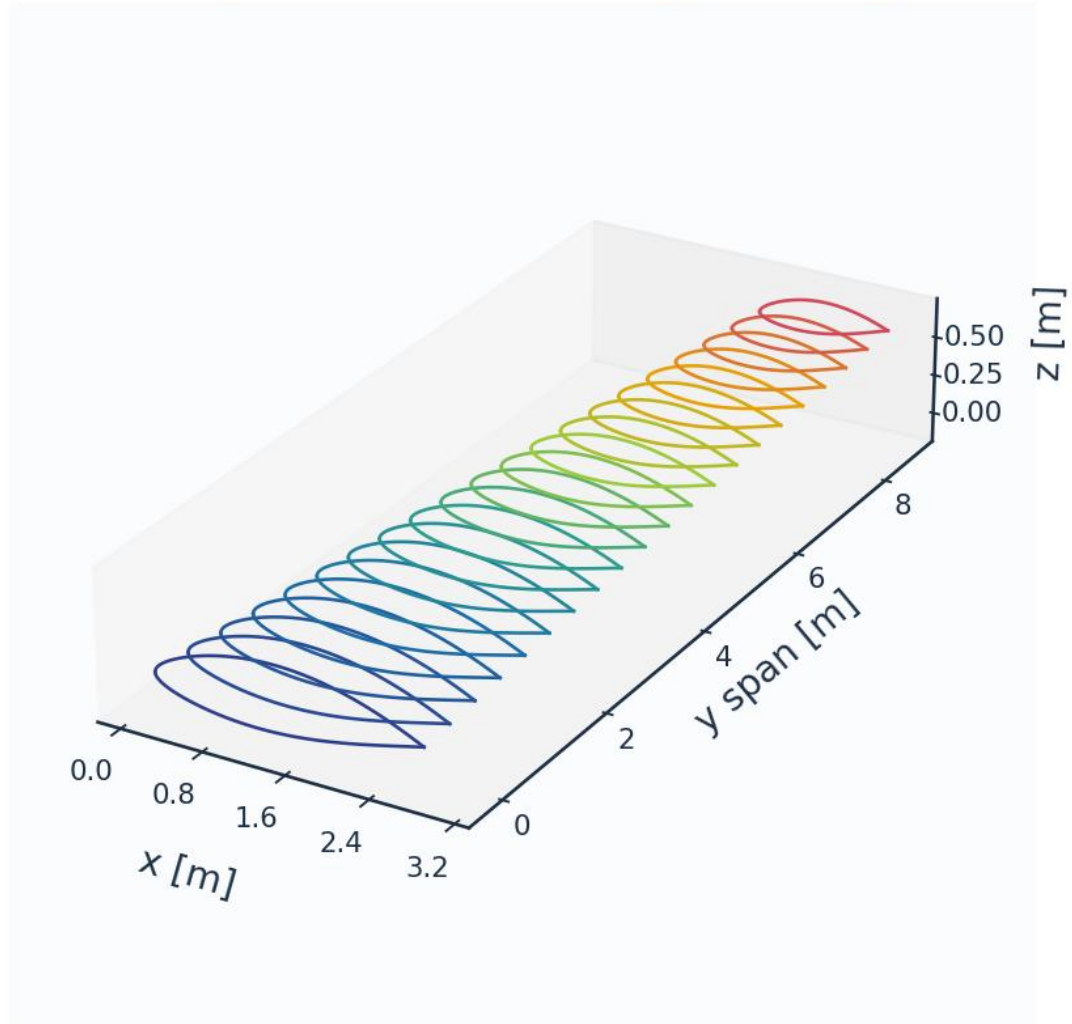


Fig. 8c. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

The surface is discretised with cosine-clustered streamwise nodes; a wall-normal reconstruction grid (first-cell $y^+ \approx 1$) supports boundary-layer profile recovery. A mesh-independence study confirms convergence.

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y_1	6.51	micron
Target wall y^+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)

LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_{τ} (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Table 9. Mesh metrics.

n_surface_panels	Cl	Cd	x_tr_upper_c	dCd_pct
120	0.5146538348994995	0.0048563824847794	0.4597401104557821	
180	0.5161840825332155	0.004944710805703	0.4729211803612302	1.8188089838549135
260	0.5170864702757358	0.0049604782218882	0.4810473518921733	0.318874385273693
360	0.5176435398599823	0.0050523311104591	0.4773197685571159	1.8516942210450529
480	0.5180064091537756	0.0051048617353106	0.4762184231006874	1.0397304472528734

Table 10. Mesh-independence study.

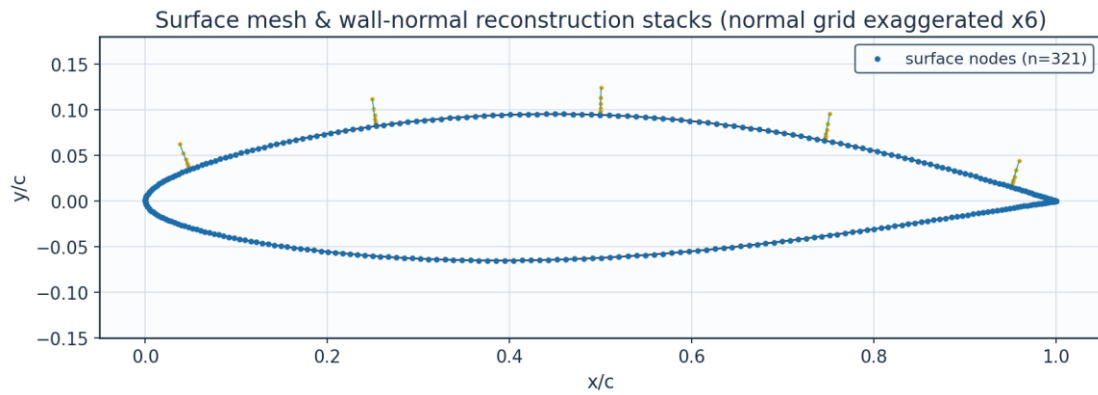


Fig. 9. Surface mesh and wall-normal stacks.

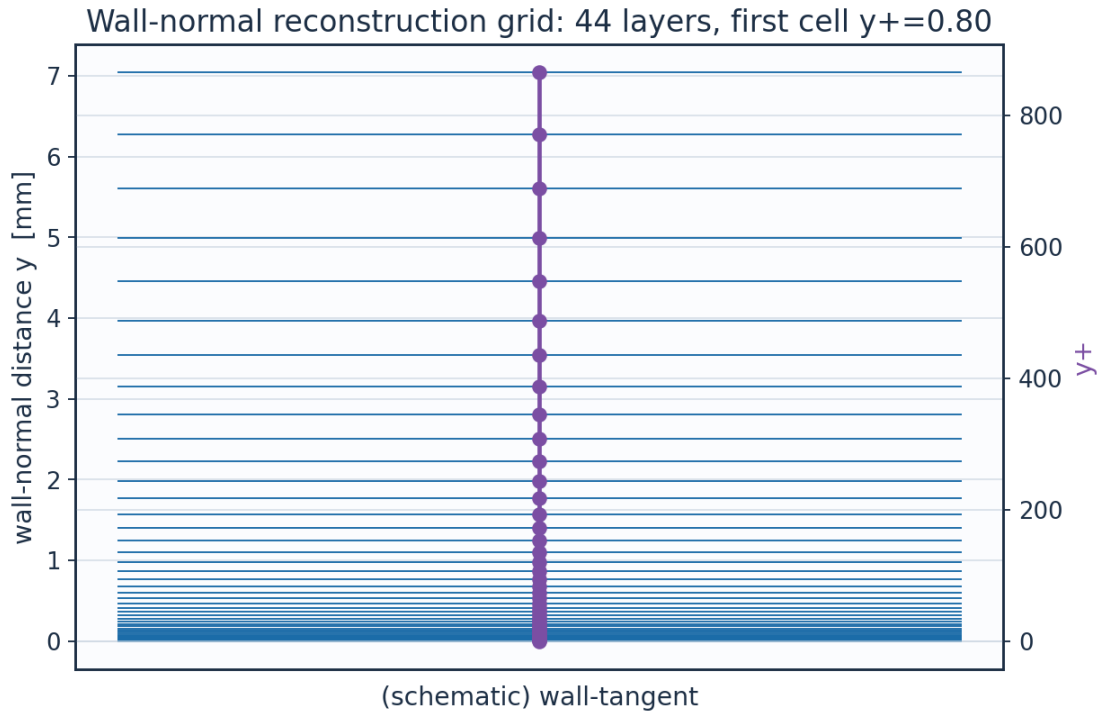


Fig. 10. Wall-normal reconstruction grid / y^+ .

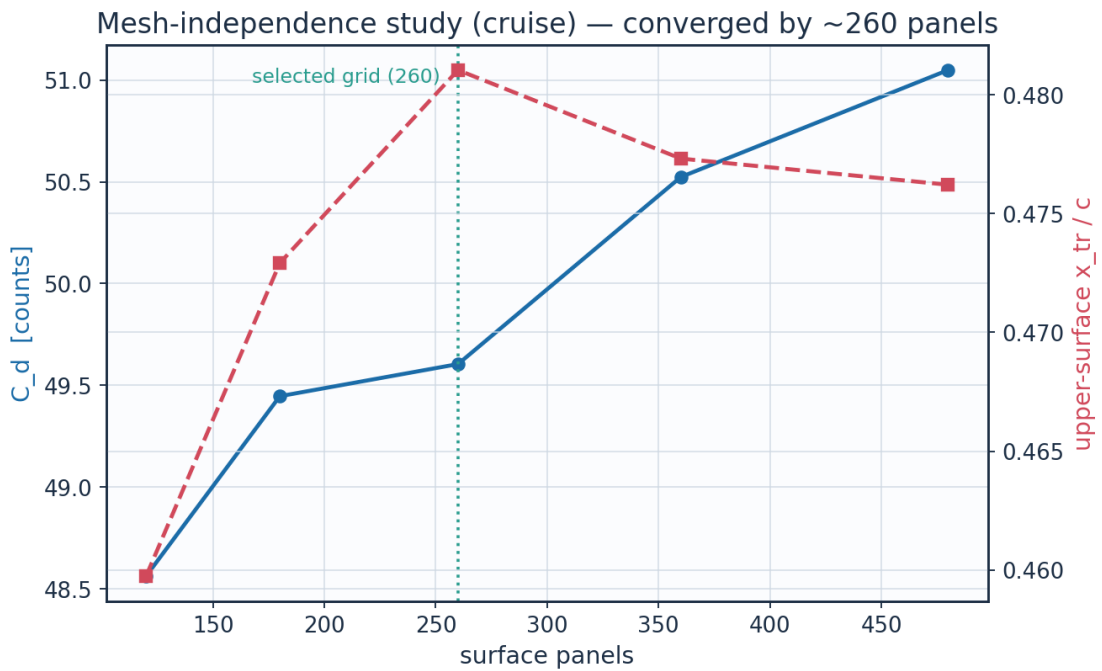


Fig. 11. Mesh-independence convergence.

layer	y_m	y_{plus}	cell_dy_m	growth_ratio
0.0	0.0	0.0	6.5127391337287455e-06	1.0
2.0	1.380700696350494e-	1.696	7.731923899562769e-	1.12

	05		06	
4.0	3.112651649852554e-05	3.8234624	9.698925339611536e-06	1.1199999999999997
6.0	5.2852109259255385e-05	6.4921512345600005	1.2166331946008715e-05	1.12
8.0	8.010469281831492e-05	9.839754508632067	1.526144679307334e-05	1.1199999999999997
10.0	0.0001142903336347	14.038988055628067	1.9143958857231197e-05	1.1199999999999997
12.0	0.0001571728014749	19.30650661697985	2.401418199051081e-05	1.12
14.0	0.0002109645691337	25.91408190033953	3.012338988889677e-05	1.12
16.0	0.0002784409624848	34.20262433578592	3.778678027663213e-05	1.1200000000000001
18.0	0.0003630833503045	44.599771966809854	4.739973717900734e-05	1.1199999999999997
20.0	0.0004692587615855	57.64195395516631	5.945823031734682e-05	1.12
22.0	0.0006024451974963	74.00206704136062	7.458440411007986e-05	1.12
24.0	0.0007695142627029	94.52419289668278	9.355867651568416e-05	1.12
26.0	0.000979085698098	120.26714756959888	0.0001173600038212	1.1200000000000006
28.0	0.0012419721066577	152.55910991130486	0.0001472163887934	1.1200000000000008
30.0	0.0015717368175549	193.0661474727408	0.0001846682381024	1.1199999999999997
32.0	0.0019853936709044	243.87817538980616	0.0002316478378757	1.1200000000000003
34.0	0.002504284827746	307.6167832089729	0.0002905790478312	1.1199999999999999
36.0	0.0031551818948881	387.57049285733575	0.0003645023575995	1.12
38.0	0.0039716671759111	487.864426240242	0.0004572317573729	1.12
40.0	0.0049958663124264	613.6731362757596	0.0005735515164485	1.1200000000000014
42.0	0.0062806217092712	771.4875821443129	0.000719463022233	1.1199999999999997

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled to 22 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism	laminar_run_pct	Cf_te
CRUISE	upper	0.505	0.481	3242000.0	1252.6	TS-natural	48.1	0.0002
CRUISE	lower	0.539	0.531	3460000.0	1161.7	TS-natural	53.1	0.00024
CLIMB	upper	0.047	0.025	506900.0	500.1	bypass	2.5	0.00014
CLIMB	lower	0.159	0.157	1706000.0	597.8	bypass	15.7	0.00019

Table 12. Transition prediction summary (transition_summary.csv).

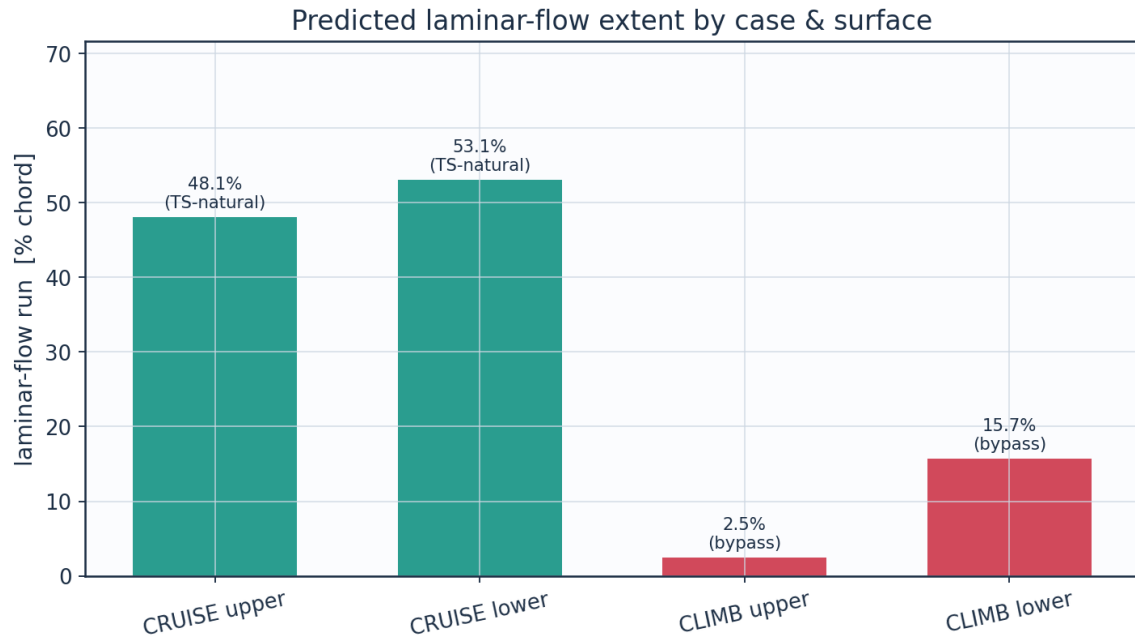


Fig. 12a. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00496	-
Section Cd (counts)	49.6	counts
Section L/D	104.2	-
Wing lift (3D est, e=0.85)	43114.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Table 13. Integrated forces.

configuration	Cd_counts	mean_laminar_pct
NLF (UTSS predicted transition)	49.6	52.2
Fully turbulent (LE trip)	87.7	0.0
Viscous drag reduction	43.4	

Table 14. NLF vs fully-turbulent drag.

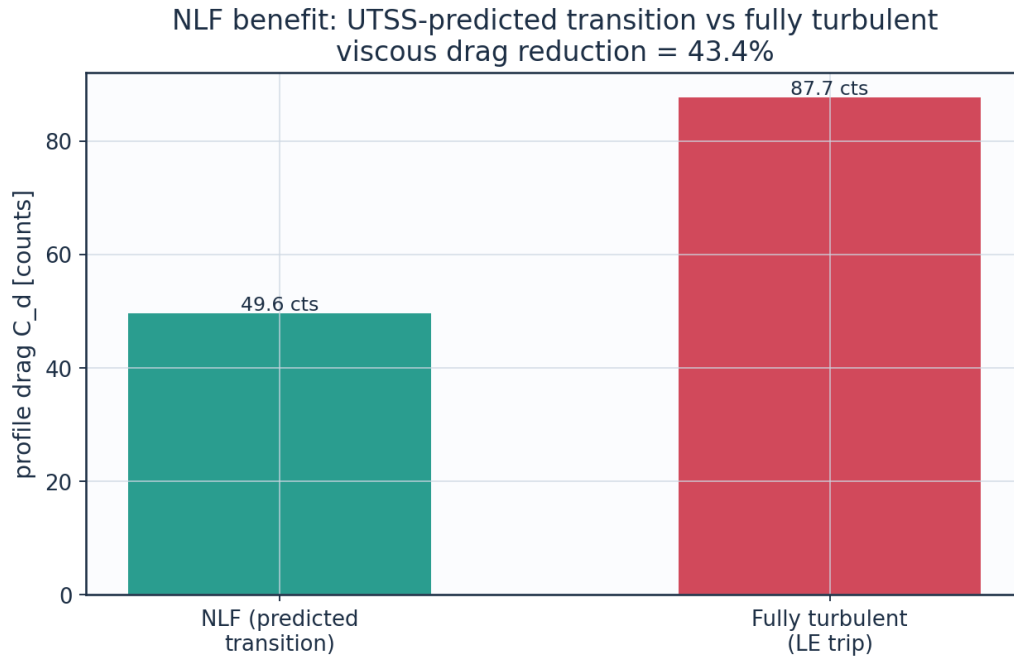


Fig. 12. Drag benefit of predicted laminar flow vs fully-turbulent.

9.3 Surface distributions — cruise

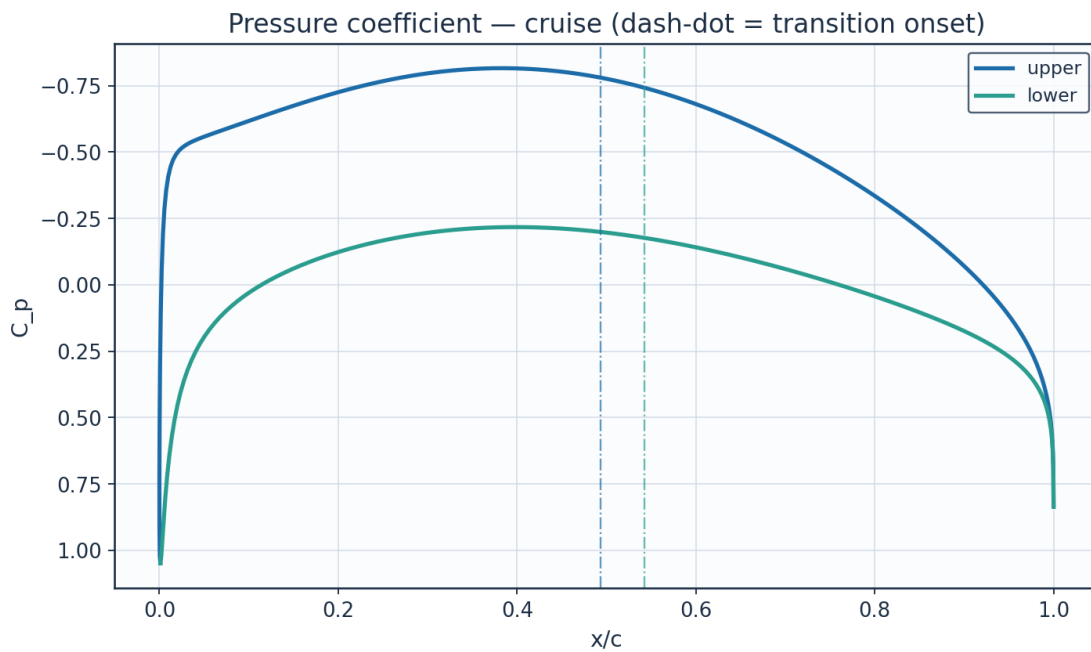


Fig. 13. Pressure coefficient C_p — cruise.

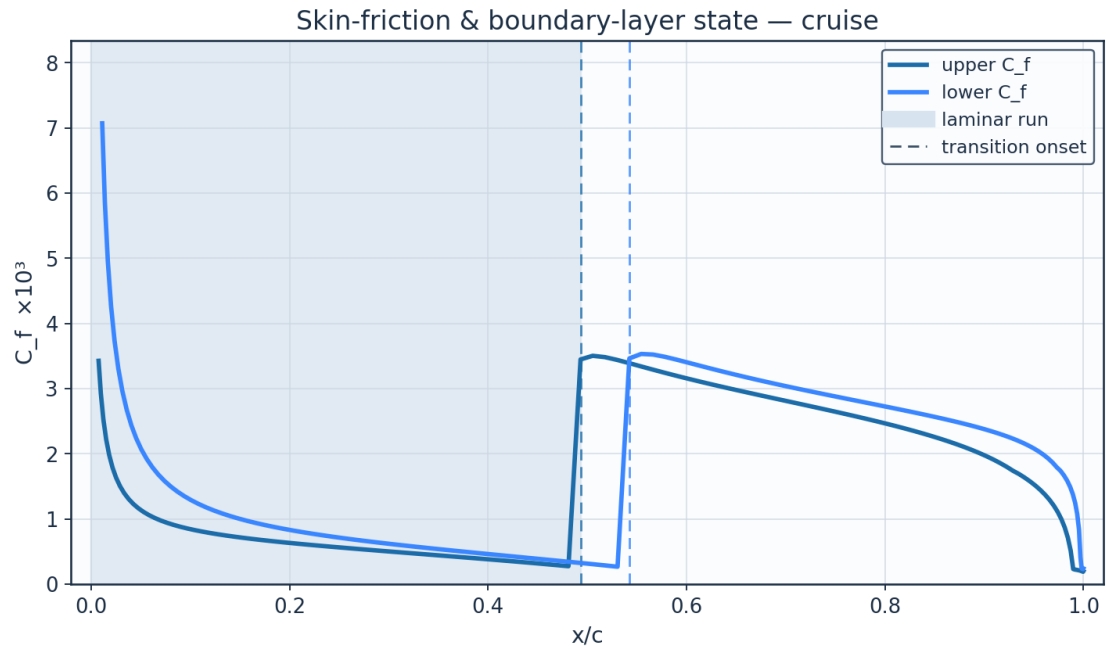


Fig. 14. Skin-friction C_f and laminar run — cruise.

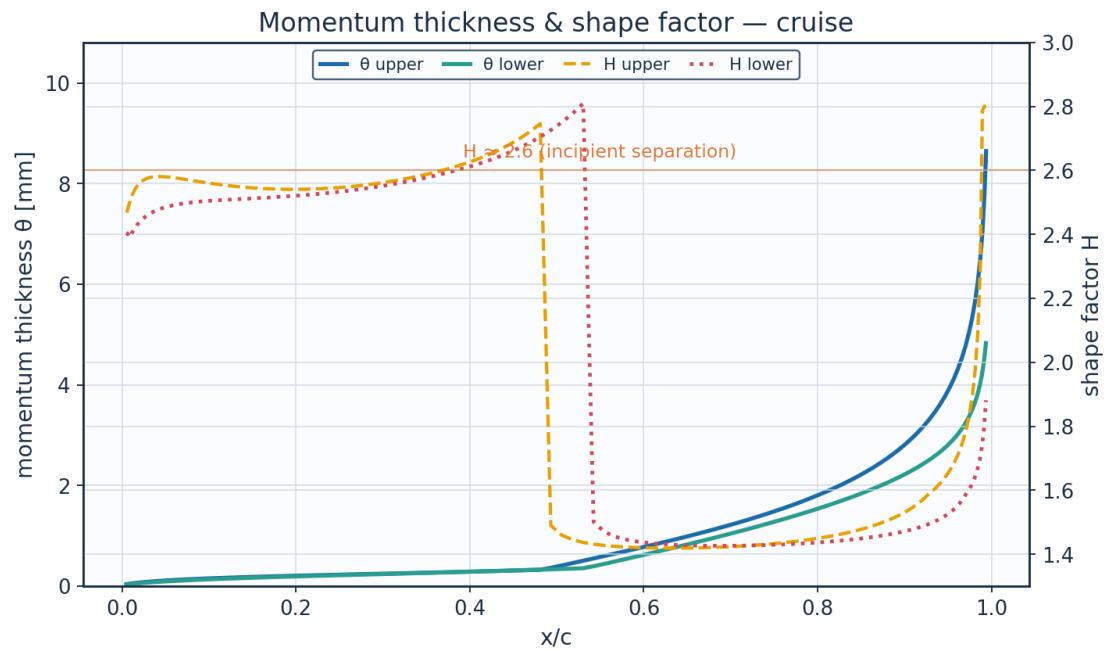


Fig. 15. Momentum thickness θ and shape factor H — cruise.

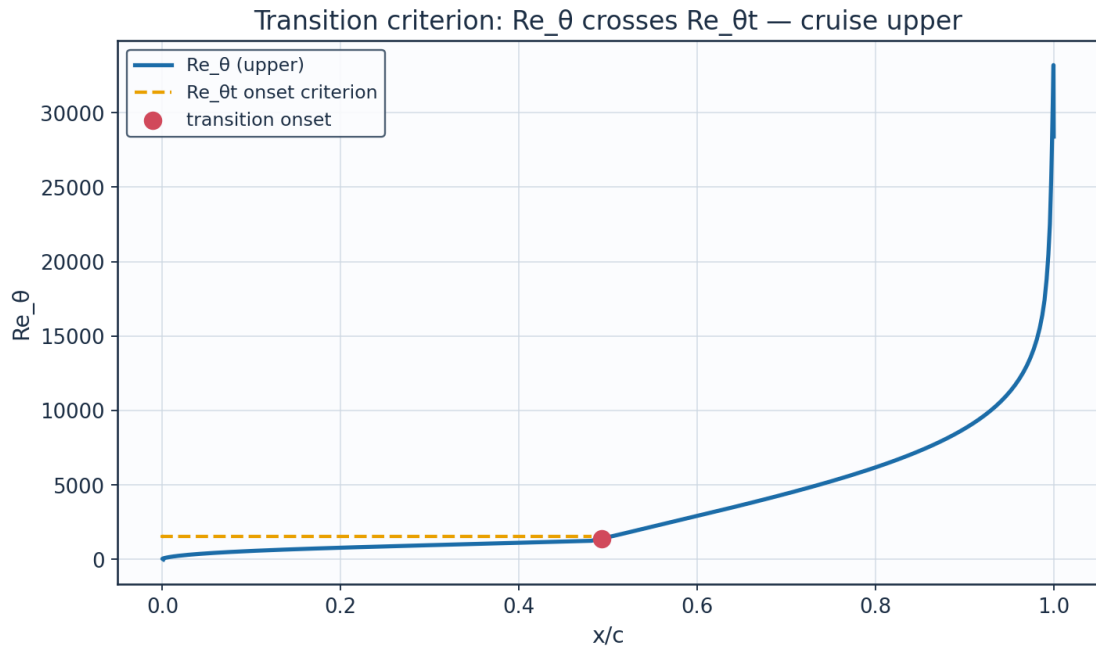


Fig. 16. Transition criterion Re_θ vs $Re_{\theta t}$ — cruise.

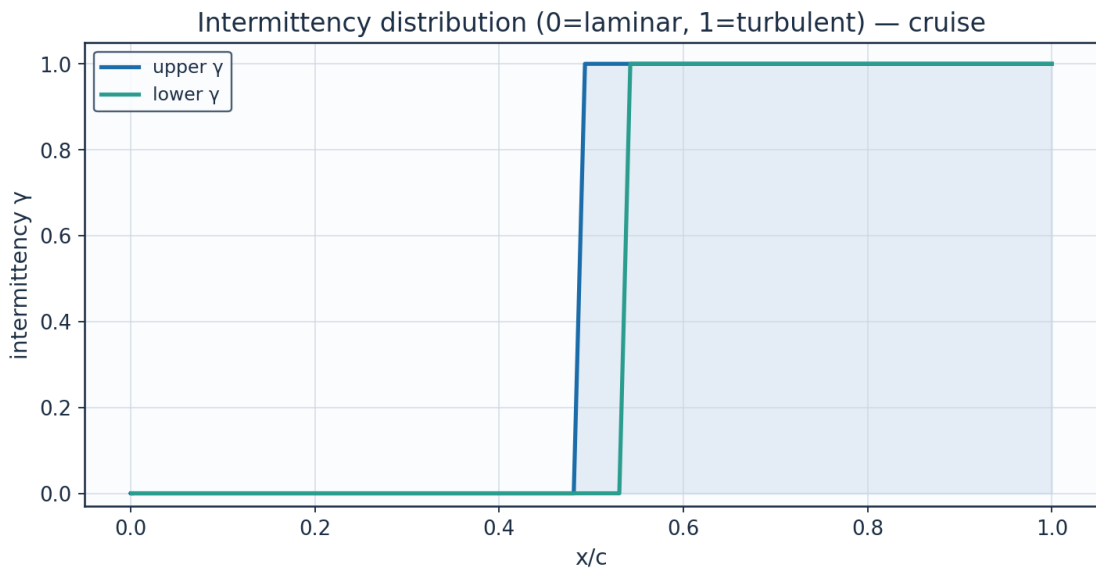


Fig. 17. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011523200325028	0.0	0.0	1.0484500005807595	0.6743779759375519	0.0054416302825443	1e-05	2.6099999414507185	2549.4326280728515	0.00017258743933047386	1569.314895	0.0	laminar
0.000242727	0.014097598	44712.17719	0.531909959	88.07774049	0.710709004	0.023894283	2.406649053	0.011615634	52.77580090	1569.314895	0.0	lami

1046013	6799438	636864	2569656	458455	4378861	3937295	5199807	1804158	88081	047386		na r
0.0039984634929024	0.0303574508340593	96282.19328257376	-0.1906595115746616	134.13696372507842	1.082365958215552	0.0359582328682185	2.4476972670491866	0.0049033697545502	117.79407830712654	1569.314895047386	0.0	la mi na r
0.0123828259304446	0.052871121292092	167686.92296127492	-0.4431856098056879	146.32725910183538	1.1807307959357134	0.0568887650828901	2.526799032644974	0.0025311224062595	201.5235379310988	1569.314895047386	0.0	la mi na r
0.0253172774881581	0.0831882375254351	263841.19035694306	-0.5161572579089404	149.61652550936128	1.2072722494368673	0.080352211592022	2.570435671814951	0.0016300084707067	290.3168275864245	1569.3148950473862	0.0	la mi na r
0.0426830314241258	0.1217109964051874	386020.6097196828	-0.5481704273505028	151.03025162861496	1.2186797614497848	0.103772113056008	2.580583184458251	0.0012290978765792	378.0669131286277	1569.3148950473862	0.0	la mi na r
0.0643227586523211	0.1683869080802935	534058.7033695233	-0.5751048472060674	152.20640645138116	1.2281702844634217	0.1261226248295291	2.5751180978540456	0.001014319469378	462.6528166644373	1569.3148950473862	0.0	la mi na r
0.0900422957300214	0.2229351474645065	707064.8018171047	-0.6054032741270788	153.51532585711524	1.2387320995430888	0.147061896549065	2.564161963996075	0.0008803325061023	543.5487097358957	1569.314895047386	0.0	la mi na r
0.1196122456699886	0.2849313998941607	903693.1416555132	-0.6399761678473315	154.99111243577192	1.250640384249193	0.166595215229562	2.5530060519809625	0.0007857599934005	620.944098845206	1569.314895047386	0.0	la mi na r
0.1527694018013956	0.353848290318842	1122271.0914502463	-0.6773382556320362	156.56525350268146	1.2633423021710868	0.184928449395984	2.544586071079718	0.0007118476071508	695.4091963616465	1569.314895047386	0.0	la mi na r
0.1892179872734863	0.4290798210019653	1360876.6587546526	-0.7150952205439175	158.13484523358554	1.2760075109988878	0.2023916578103691	2.5406811264258966	0.0006490800889309	767.7434764607328	1569.314895047386	0.0	la mi na r
0.2286307839316745	0.5099588956971477	1617394.0707776232	-0.7504707657473884	159.5866938991024	1.2877226380431313	0.2193922365608963	2.542537498057767	0.0005921602196781	838.8898395967217	1569.314895047386	0.0	la mi na r
0.2706503108546585	0.59577083610005	1889556.6398858647	-0.7805852261678627	160.80872570370784	1.2975833474841956	0.2363836789268968	2.5512529226949856	0.0005378677494562	909.876344635952	1569.3148950473858	0.0	la mi na r
0.3148902862023041	0.6857636986922375	2174979.491676561	-0.80268469982041	161.69756919660222	1.3047555236820498	0.2538448491237488	2.5680664381414267	0.0004840364730578	981.7732478885148	1569.314895047386	0.0	la mi na r

			01									
0.3609	0.7791	24711	-	162.16	1.3085	0.2722	2.5946	0.0004	1055.6	1569.3	0.0	la
37646	55626	82.873	0.8143	46593	24525	67502	31894	29002	61545	14895		mi
56271	03463	46131	64452	46477	62892	51022	74832	65750	95283	04738		na
2	46	45	32521	78	9	08	6	13	9	6		r
			26									
0.4083	0.8751	27756	-	162.14	1.3083	0.2921	2.6342	0.0003	1132.6	1569.3	0.0	la
55390	39546	07.063	0.8137	11197	34582	50687	17595	71260	12033	14895		mi
17306	45295	36620	74739	75158	23116	73575	76992	97806	04376	04738		na
84	7	6	53945	5	16	02		25	86	6		r
			51									
0.4566	0.9728	30856	-	161.58	1.3038	0.3140	2.6980	0.0003	1213.6	1569.3	0.0	la
86447	85870	20.921	0.7997	10430	15261	01047	20126	09051	74874	14895		mi
36747	56001	96654	78064	77768	57448	66910	44801	27946	88467	04738		na
44	44	44	70728	75	34	76	7	61	65	6		r
			27									
0.5054	1.0715	33985	-	160.46	1.2947	0.4098	1.4600	0.0035	1570.3		1.0	tur
58667	44176	27.236	0.7720	27537	91661	72404	43706	03471	48515			bu
99852	36863	81889	27910	07129	27754	82188	41265	05008	91842			le
56	5	24	22110	7	76	79	77	35	3			nt
			03									
0.5541	1.1702	37115	-	158.78	1.2812	0.5971	1.4274	0.0033	2267.1		1.0	tur
90867	45063	68.604	0.7309	78833	76947	28617	52330	40841	07989			bu
66523	74149	06482	53208	16317	32902	25975	27368	52902	07942			le
94	47	8	29388	2	13	82	23	09	7			nt
			42									
0.6023	1.2681	40219	-	156.57	1.2634	0.7881	1.4198	0.0031	2956.0		1.0	tur
99717	03309	37.436	0.6776	84251	48585	25994	01865	48892	73192			bu
62090	91279	53568	53007	17026	38027	18448	78749	05432	63800			le
82	1	5	07654	46	68	23	42	37	8			nt
			81									
0.6496	1.3642	43267	-	153.87	1.2416	0.9864	1.4190	0.0029	3644.0		1.0	tur
07126	23169	92.774	0.6137	22197	11915	40805	82609	80521	58542			bu
44050	95281	79366	23995	38402	64231	29800	16054	27065	09416			le
3	22	8	79041	4	29	96	82	61	98			nt
			99									
0.6953	1.4577	46232	-	150.71	1.2161	1.1967	1.4227	0.0028	4341.3		1.0	tur
47671	06209	85.129	0.5410	74710	55900	42632	82822	27513	55658			bu
73678	48466	50581	53727	46898	43120	42598	06791	94568	75055			le
39	56	1	68791	48	1	84	26	42	9			nt
			76									
0.7391	1.5476	49085	-	147.16	1.1875	1.4238	1.4303	0.0026	5057.7		1.0	tur
75615	61543	88.956	0.4616	68703	05712	43505	80566	82115	05642			bu
76998	81474	03955	12518	97202	10061	52007	31041	84481	05401			le
33	4	5	26452	25	74	2	52	86				nt
			6									
0.7806	1.6332	51799	-	143.27	1.1560	1.6731	1.4420	0.0025	5803.3		1.0	tur
71085	17905	40.926	0.3772	17466	75529	21701	70085	38319	21863			bu
14247	01870	63225	69065	98238	27907	21456	55512	25804	35687			le
59	1	7	09069	22	52	56	32	26	8			nt
			94									
0.8194	1.7135	54346	-	139.07	1.1222	1.9511	1.4585	0.0023	6590.3		1.0	tur
45102	36656	81.206	0.2896	64000	22814	67862	97550	90774	41500			bu
96803	13103	41017	42823	66341	49229	12891	82123	66618	85127			le
74	57	8	00443	6	63	9	63	39	9			nt
			23									
0.8551	1.7878	56702	-	134.61	1.0862	2.2667	1.4814	0.0022	7434.8		1.0	tur
43310	24762	94.594	0.1999	24371	02604	81389	27958	33706	39161			bu
57486	77398	42127	93692	09550	94725	97946	14021	81186	57684			le
8	5		84881	54	66	9	12	22				nt
			82									

0.8874 48376 11318 26	1.8553 46811 45599 4	58844 48.641 06943 6	- 0.1091 38200 00321 98	129.89 24515 20186 65	1.0481 16520 53658 5	2.6326 82231 13199 8	1.5132 64092 42162 9	0.0020 59633 02132 82	8360.0 66367 99245 8		1.0	tur bu le nt
0.9160 81229 91338 4	1.9154 35385 45516 16	60750 26.556 43811 1	- 0.0173 69522 59207 68	124.90 15687 55567 9	1.0078 44536 93447 95	3.0687 54484 38882 3	1.5593 51150 17529 32	0.0018 57236 86846 04	9402.3 80056 07598 5		1.0	tur bu le nt
0.9408 01367 76385 24	1.9674 99407 22596 13	62401 53.669 20543 7	0.0756 60067 35794 82	119.58 35624 55574 66	0.9649 32957 43846 64	3.6087 95020 40960 7	1.6205 58245 59858 72	0.0016 33651 67167 32	10623. 26564 01802 63		1.0	tur bu le nt
0.9614 06511 28321 76	2.0110 30361 48001 73	63782 17.163 87019 8	0.1711 81141 30624 9	113.81 34551 80805 52	0.9183 73325 30299 52	4.3146 71936 62309 2	1.7101 72307 66575 11	0.0013 70700 02601 06	12132. 20616 28393 1		1.0	tur bu le nt

Table 15. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm	H_shape	Cf	Re_theta	Re_theta_trans	intermittency_gamma	state
0.0011 52320 03250 28	0.0	0.0	1.0484 50000 58075 95	0.6743 77975 93755 19	0.0054 41630 28254 43	1e-05	2.6099 99969 21188 65	2549.4 32493 38546 4	0.0001 72587 43933 07	1569.3 14895 04738 6	0.0	la mi na r
0.0067 20774 97236 91	0.0175 58980 96768 28	55690. 35452 34006 45	0.7967 88059 19874 23	61.869 95831 42874 79	0.4992 35518 88645 79	0.0361 89332 95949 38	2.4014 55110 01165 2	0.0108 85848 95217 75	56.727 82683 07835 6	1569.3 14895 04738 6	0.0	la mi na r
0.0168 97481 50526 17	0.0419 27102 59539 91	13297 6.6922 10909	0.5091 25482 40715 59	89.950 65248 46617 9	0.7258 21737 89804 35	0.0532 88056 07865 99	2.4275 91205 30259 14	0.0049 53563 39743 33	120.09 68285 24372 28	1569.3 14895 04738 6	0.0	la mi na r
0.0315 87190 34084 9	0.0742 42760 85277 24	23546 9.5685 52791 4	0.3201 16148 47371 03	104.06 54470 09191 12	0.8397 15572 00463 43	0.0736 10231 64887 2	2.4636 43616 82539 2	0.0029 59051 22110 7	190.56 80972 42451 16	1569.3 14895 04738 6	0.0	la mi na r
0.0506 50104 83468 89	0.1147 42805 66774 19	36392 0.1806 98816 6	0.1932 57759 39698 61	112.43 01677 32619 13	0.9072 11426 28479 7	0.0947 41237 37917 27	2.4852 76623 75725 3	0.0020 67873 84466 46	263.74 82607 82428 25	1569.3 14895 04738 62	0.0	la mi na r
0.0739 02667 92458 39	0.1632 86856 80577 34	51788 3.2963 74761 5	0.1005 40820 64840 53	118.11 26263 41393 7	0.9530 63811 66530 09	0.1158 16630 73062 97	2.4975 92677 31733 6	0.0015 84548 42350 6	337.57 02582 48979 14	1569.3 14895 04738 6	0.0	la mi na r
0.1011 19142 10955 74	0.2195 30525 04901 36	69626 6.6450 39893	0.0275 01103 73430 19	122.37 11244 77975 4	0.9874 26102 91000 72	0.1364 60146 27314 26	2.5047 42863 77285 5	0.0012 86665 40794 98	410.99 25360 51369 2	1569.3 14895 04738 62	0.0	la mi na r
0.1320 34052 45495 44	0.2829 94629 23130 39	89755 0.4477 80308 7	- 0.0329 49480 64718 06	125.76 58705 52816 2	1.0148 18683 48282 34	0.1565 58134 49932 62	2.5096 84230 40560 53	0.0010 84900 82792 33	483.55 37925 46694 8	1569.3 14895 04738 58	0.0	la mi na r
0.1663	0.3531	11199	-	128.55	1.0373	0.1761	2.5146	0.0009	555.17	1569.3	0.0	la

45511 63160 67	01330 53650 2	02.021 44727 8	0.0840 89705 79544 8	35542 38377 76	12810 66783	69248 74030 36	15367 27199 8	37398 92721 42	25110 74065 9	14895 04738 56		mi na r
0.2037 19371 58191 91	0.4291 98466 63705 09	13612 52.957 21369 96	- 0.1271 71832 79815 87	130.84 60216 58987 44	1.0558 10983 18064 53	0.1954 74097 67175	2.5212 76076 95081 6	0.0008 22259 72275 85	626.03 60238 74914 8	1569.3 14895 04738 6	0.0	la mi na r
0.2437 94056 30589 38	0.5105 78867 09950 84	16193 60.381 63382 23	- 0.1623 62939 49216 68	132.68 26339 36490 7	1.0706 30810 25560 3	0.2147 40279 68264 12	2.5311 47244 29188 65	0.0007 27032 07815 97	696.52 55180 97977 6	1569.3 14895 04738 6	0.0	la mi na r
0.2861 85839 50534 81	0.5964 96549 11452 82	18918 58.323 28838 8	- 0.1893 00550 93779 56	134.06 75651 33079 26	1.0818 05972 86099 4	0.2342 94875 32804 18	2.5456 04320 67108 34	0.0006 44151 27350 72	767.15 63975 09766 7	1569.3 14895 04738 6	0.0	la mi na r
0.3304 94258 79381 13	0.6861 79937 75148 77	21762 99.642 36291 36	- 0.2074 61412 92644 59	134.99 13376 06891 08	1.0892 59994 85909 8	0.2545 01290 30128 58	2.5660 50106 71393 87	0.0005 68751 54370 33	838.52 53597 85778	1569.3 14895 04738 6	0.0	la mi na r
0.3763 07324 00294 27	0.7788 41767 61081 37	24701 87.434 89202	- 0.2164 23488 27837 99	135.44 43196 35477	1.0929 15156 81893 9	0.2757 38837 84056 22	2.5940 40102 98217 05	0.0004 97528 66615 05	911.26 04426 37806 4	1569.3 14895 04738 62	0.0	la mi na r
0.4232 06194 57674 89	0.8736 85455 08345 47	27709 95.243 12299 02	- 0.2160 46718 78413 36	135.42 53140 60304 83	1.0927 61798 73625 5	0.2983 84063 78440 28	2.6321 61672 83081 06	0.0004 28108 15109 95	985.97 29363 54926 8	1569.3 14895 04738 62	0.0	la mi na r
0.4707 69074 33519 84	0.9699 08166 57242 24	30761 76.786 73797 05	- 0.2065 75740 70034 36	134.94 64691 14026 26	1.0888 97946 04045 75	0.3227 93347 50536 68	2.6887 22664 98481 06	0.0003 58551 16268 47	1063.2 12108 14244 9	1569.3 14895 04738 6	0.0	la mi na r
0.5185 74193 40729 35	1.0667 01325 94151 12	33831 67.572 28672 46	- 0.1886 63903 91346 69	134.03 50378 60133	1.0815 43506 70731 1	0.3492 87011 22002 63	2.7783 12019 91745 67	0.0002 86440 52963 89	1143.4 25571 99893 6	1569.3 14895 04738 58	0.0	la mi na r
0.5662 01898 02471 88	1.1632 49726 11984 71	36893 82.075 53692 4	- 0.1633 24624 77316 74	132.73 23845 84628 68	1.0710 32253 72372 25	0.4876 70102 29291 98	1.4555 26269 63802 51	0.0035 22898 16440 39	1579.2 95721 66578 2		1.0	tur bu le nt
0.6132 36022 35407 45	1.2587 30610 74833 58	39922 10.828 80882 6	- 0.1318 26376 04350 61	131.09 07600 19421 88	1.0577 85804 00955 62	0.6763 27242 15955 64	1.4331 44032 07854 51	0.0033 52159 01007 33	2165.6 72063 90331 5		1.0	tur bu le nt
0.6592 64843 41077 2	1.3523 14020 52265 4	42890 21.519 44244 1	- 0.0955 55444 09005 39	129.16 85290 40938 6	1.0422 75110 19127 53	0.8670 77763 02855 72	1.4270 48598 72618 9	0.0031 77629 09115 11	2739.4 30441 69190 3		1.0	tur bu le nt
0.7038 81995	1.4431 65382	45771 67.201	- 0.0558	127.02 45984	1.0249 75497	1.0616 34100	1.4263 75883	0.0030 25341	3303.2 95768		1.0	tur bu

09891 86	46492 15	97440 2	72264 25655 04	3724	64010 8	60811 18	69640 1	03714 9	69136 9			le nt
0.7466 87725 12314 54	1.5304 50847 76179 95	48540 03.227 71357 7	- 0.0139 82254 57569 55	124.71 27010 62973 96	1.0063 20542 68780 44	1.2617 52524 51783 94	1.4289 73586 75092 48	0.0028 89811 17619 37	3860.5 35679 96597 67		1.0	tur bu le nt
0.7872 90817 05456 05	1.6133 45372 85086 3	51169 12.874 84830 4	0.0291 67159 17160 43	122.27 59555 95345 72	0.9866 58174 69741 42	1.4688 46937 10231 85	1.4341 82818 72347 75	0.0027 65397 57762 78	4413.4 88580 01919 6		1.0	tur bu le nt
0.8253 11381 55209 68	1.6910 43098 74663 05	53633 40.267 68853 3	0.0729 62068 85514 11	119.74 17781 22673 58	0.9662 09617 11001 26	1.6844 32286 20218 3	1.4419 14908 84011 3	0.0026 47409 49711 18	4964.5 44291 80414 4		1.0	tur bu le nt
0.8603 84569 42783 87	1.7627 69288 93695 93	55908 28.239 09553 2	0.1171 78365 82090 99	117.11 67909 03076	0.9450 28305 65689 84	1.9108 40870 47753 73	1.4525 11868 12254 86	0.0025 31296 62993 6	5517.5 98222 67949 7		1.0	tur bu le nt
0.8921 65104 53399 4	1.8277 92963 07022 2	57970 58.399 69313 5	0.1620 35480 87852 07	114.38 07835 11247 76	0.9229 51160 18674 96	2.1523 54552 85059 45	1.4668 62113 06587 72	0.0024 11646 73234 93	6080.1 43355 85604 4		1.0	tur bu le nt
0.9203 32404 83917 28	1.8854 39395 75766 85	59798 90.779 27608 8	0.2082 75008 29401 14	111.47 77474 37842 15	0.8995 26241 85901 14	2.4171 61070 62748 13	1.4867 83298 56205 24	0.0022 80807 65683 98	6666.6 56691 25225 2		1.0	tur bu le nt
0.9445 95976 40102 88	1.9351 01792 46614 69	61374 00.858 26456 7	0.2573 13910 01485 05	108.29 96644 45970 88	0.8738 81939 60408 83	2.7211 75441 89483 6	1.5160 04914 28132 94	0.0021 26349 31729 32	7304.9 33730 13811 5		1.0	tur bu le nt
0.9647 00736 13354 88	1.9762 51652 32757 04	62679 12.434 56211 6	0.3115 83506 76964 56	104.65 22734 79347 6	0.8444 50739 52869 62	3.0967 93135 86877 7	1.5630 39356 52050 54	0.0019 25035 82529 71	8050.1 59853 44564 2		1.0	tur bu le nt
0.9804 31941 29791 1	2.0084 47529 88663 1	63700 25.412 45964 3	0.3753 69091 58485 07	100.17 04201 74899 48	0.8082 86170 79481 93	3.6174 41273 62506 4	1.6360 57768 85093 63	0.0016 65906 90041 53	9023.2 44677 83018 7		1.0	tur bu le nt

Table 16. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled).

(table sampled to 30 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated T_u)

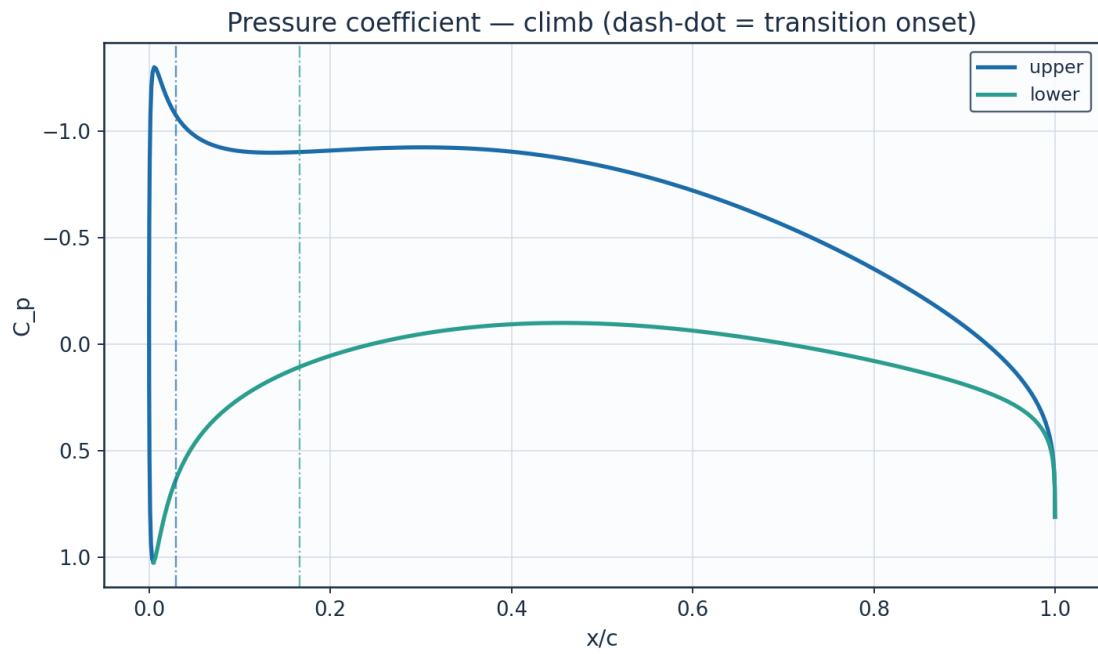


Fig. 18. Pressure coefficient C_p — climb.

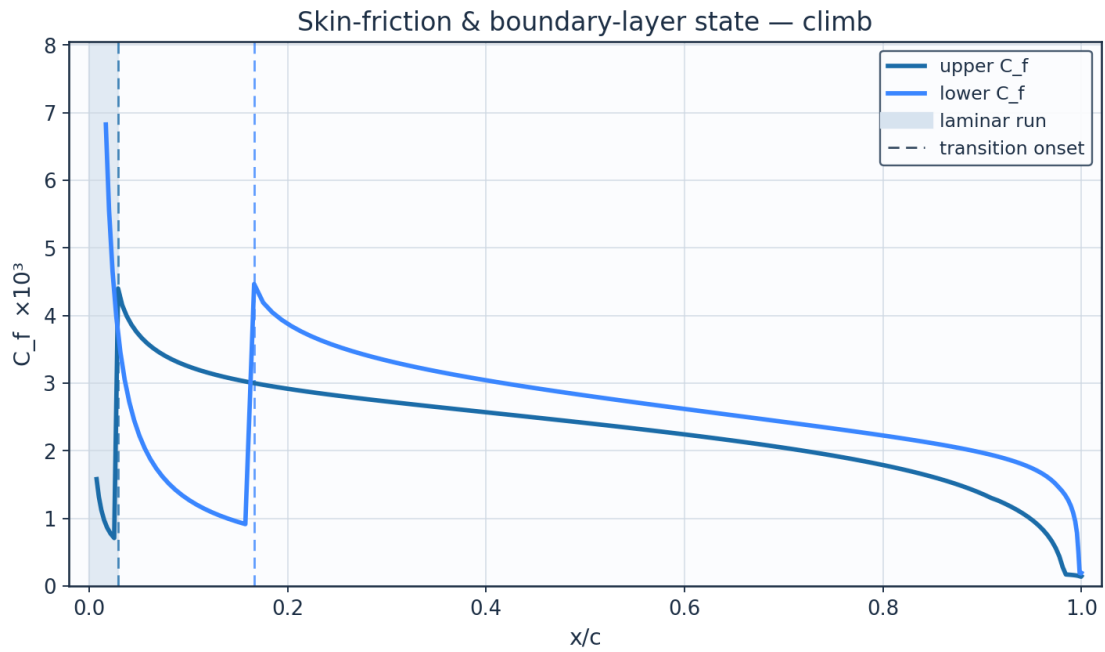


Fig. 19. Skin-friction C_f and laminar run — climb.

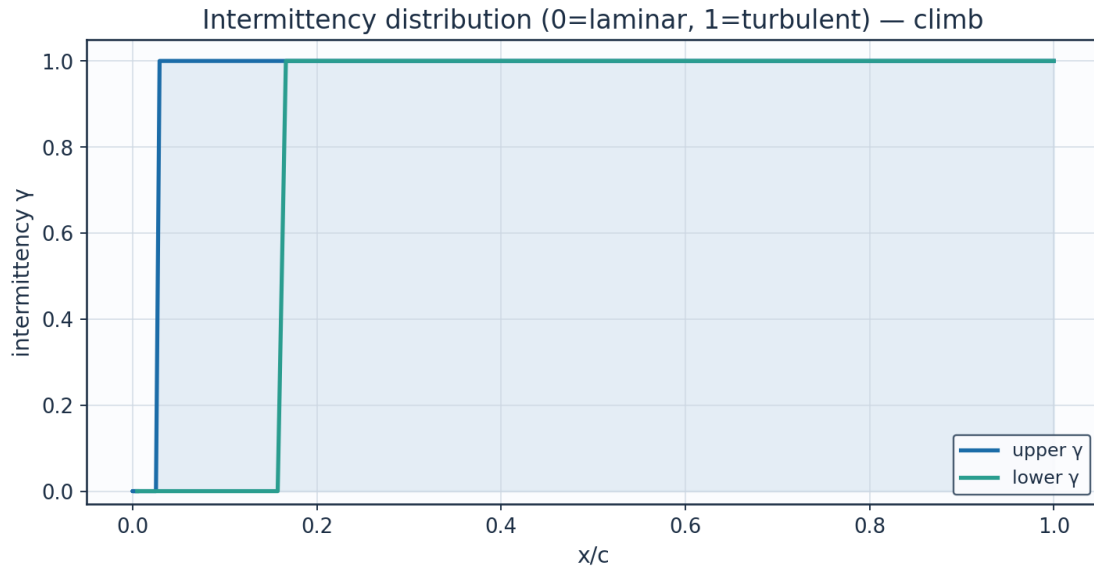


Fig. 20. Intermittency γ — climb.

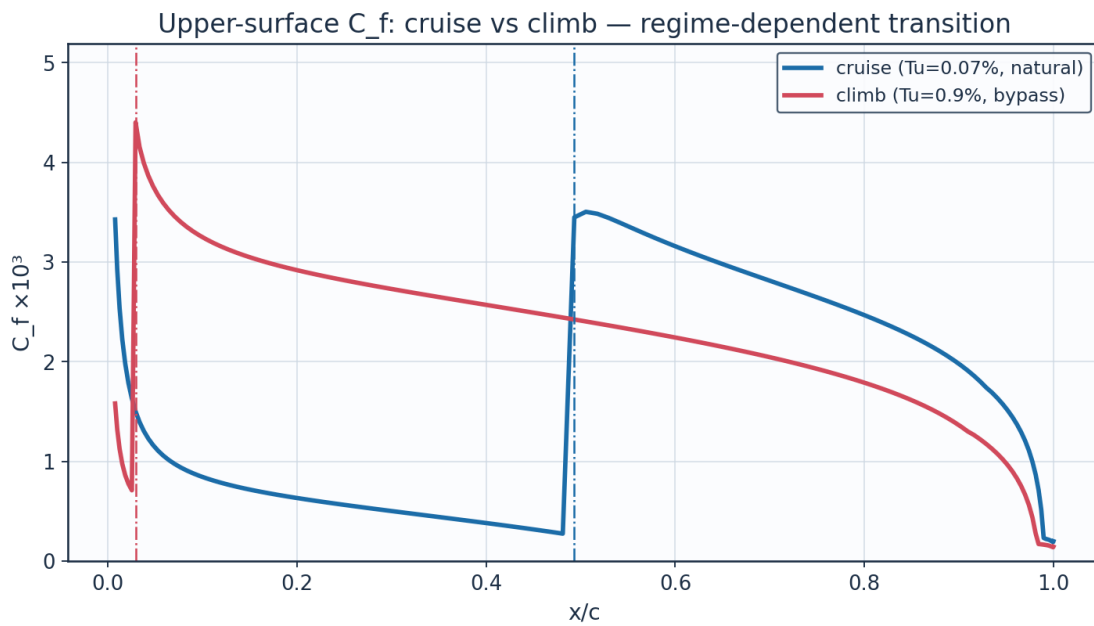


Fig. 21. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

alpha_deg	Cl	Cd	L_over_D	xtr_upper_c	xtr_lower_c
-3.0	-0.1114	0.00559	-19.9	0.614	0.223
-2.0	0.0287	0.00499	5.8	0.59	0.365
-1.0	0.1683	0.00483	34.8	0.566	0.423
0.0	0.3077	0.00484	63.5	0.53	0.471
1.0	0.4472	0.00495	90.4	0.493	0.507
2.0	0.5872	0.00506	116.0	0.457	0.554
3.0	0.7281	0.0056	130.1	0.373	0.578

4.0	0.8704	0.00749	116.2	0.09	0.613
5.0	1.0146	0.0084	120.8	0.033	0.636
6.0	1.1612	0.00897	129.5	0.029	0.671
7.0	1.3111	0.00983	133.4	0.018	0.693
8.0	1.4654	0.01074	136.5	0.015	0.715

Table 17. Aerodynamic polar (aero_polar.csv).

Aerodynamic polars (UTSS, cruise Re) — AETHER-NLF 25 section

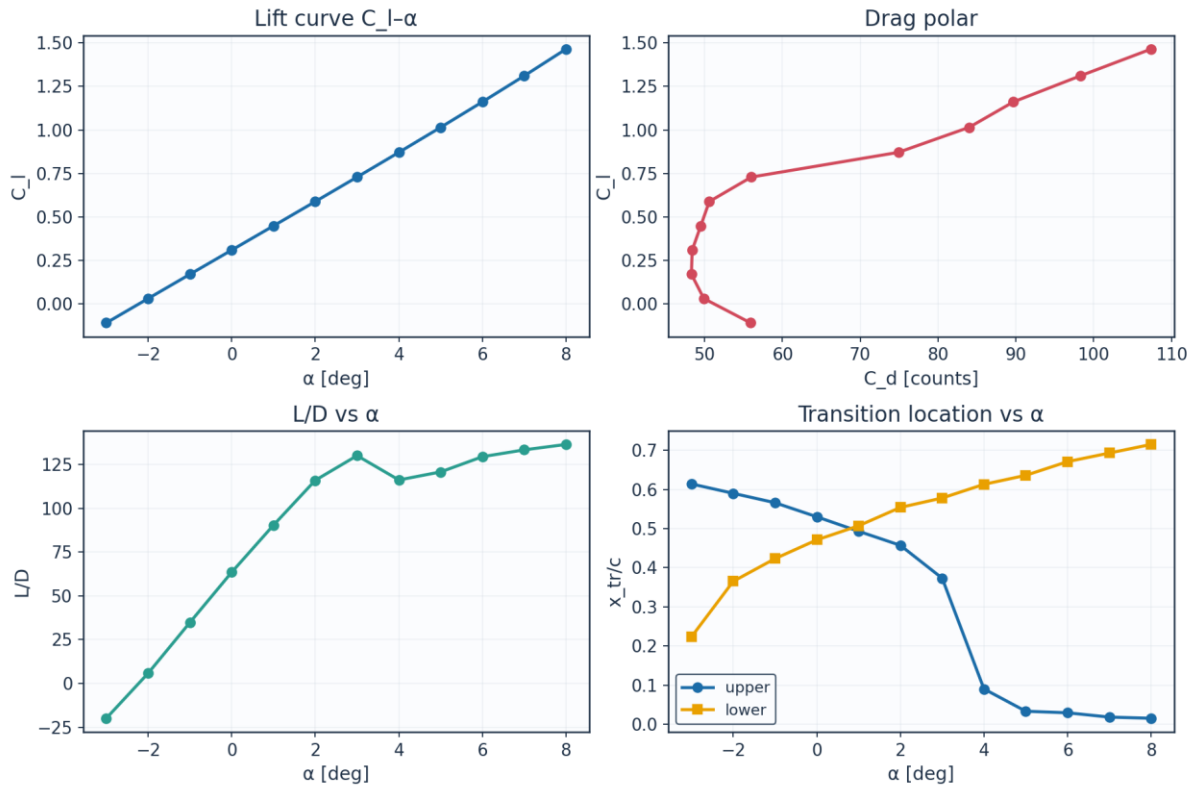


Fig. 22. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

eta	y_m	chord_m	Re_local	alpha_eff_deg	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.0	2.6	8246200.0	1.5	0.42	0.495	0.00516	0.458
0.089	0.757	2.484	7878900.0	1.23	0.457	0.495	0.00499	0.476
0.178	1.515	2.368	7511500.0	0.97	0.469	0.495	0.00496	0.482
0.267	2.272	2.253	7144200.0	0.7	0.493	0.483	0.0049	0.488
0.356	3.029	2.137	6776900.0	0.43	0.505	0.483	0.00488	0.494
0.445	3.786	2.021	6409500.0	0.16	0.53	0.483	0.0048	0.506
0.535	4.544	1.905	6042200.0	-0.1	0.542	0.471	0.00484	0.506
0.624	5.301	1.789	5674900.0	-0.37	0.566	0.471	0.00477	0.519
0.713	6.058	1.673	5307600.0	-0.64	0.578	0.471	0.00478	0.525
0.802	6.815	1.558	4940200.0	-0.91	0.59	0.471	0.0048	0.531
0.891	7.573	1.442	4572900.0	-1.17	0.602	0.459	0.00488	0.531
0.98	8.33	1.326	4205600.0	-1.44	0.614	0.459	0.00492	0.537

Table 18. Span-wise distribution (spanwise_distribution.csv).

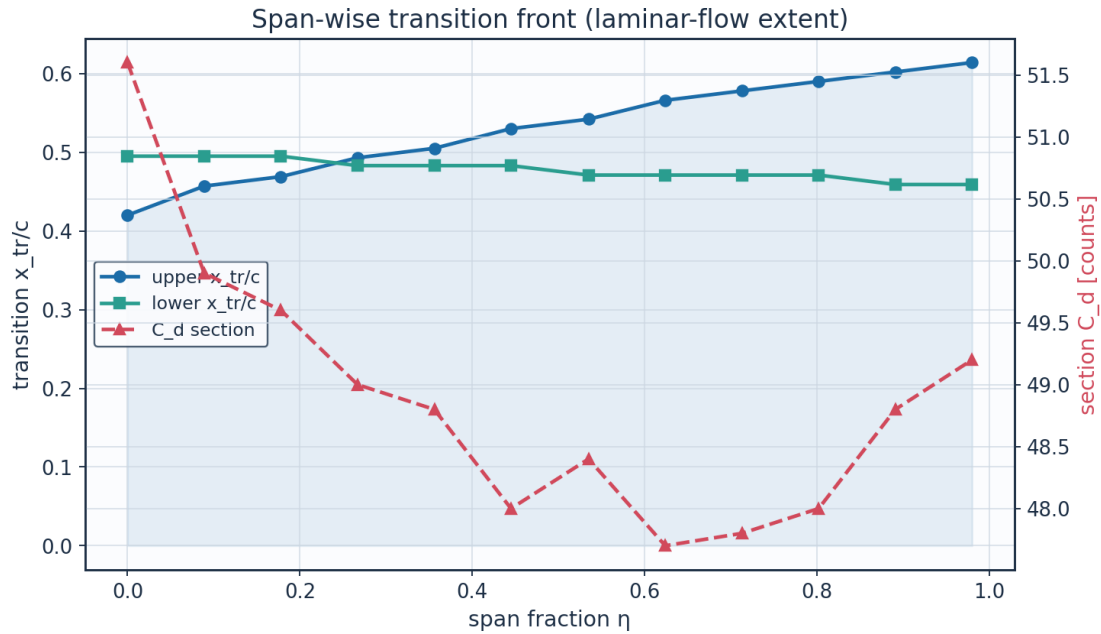


Fig. 23. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

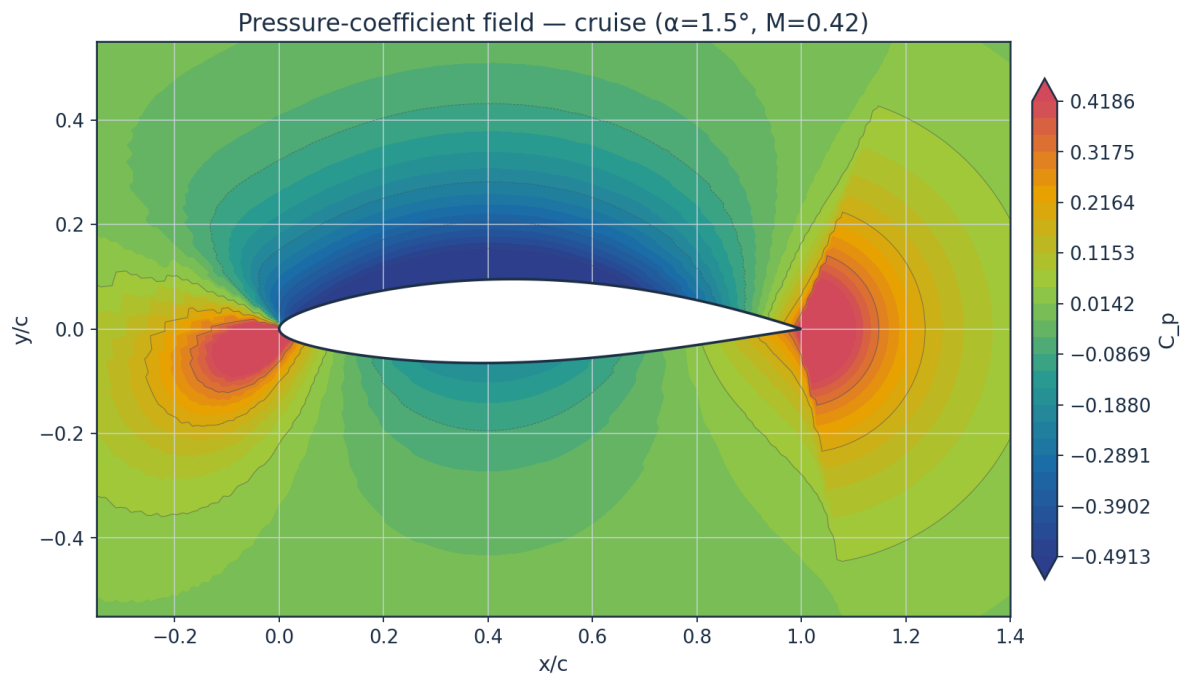


Fig. 24. Pressure-coefficient contour — cruise.

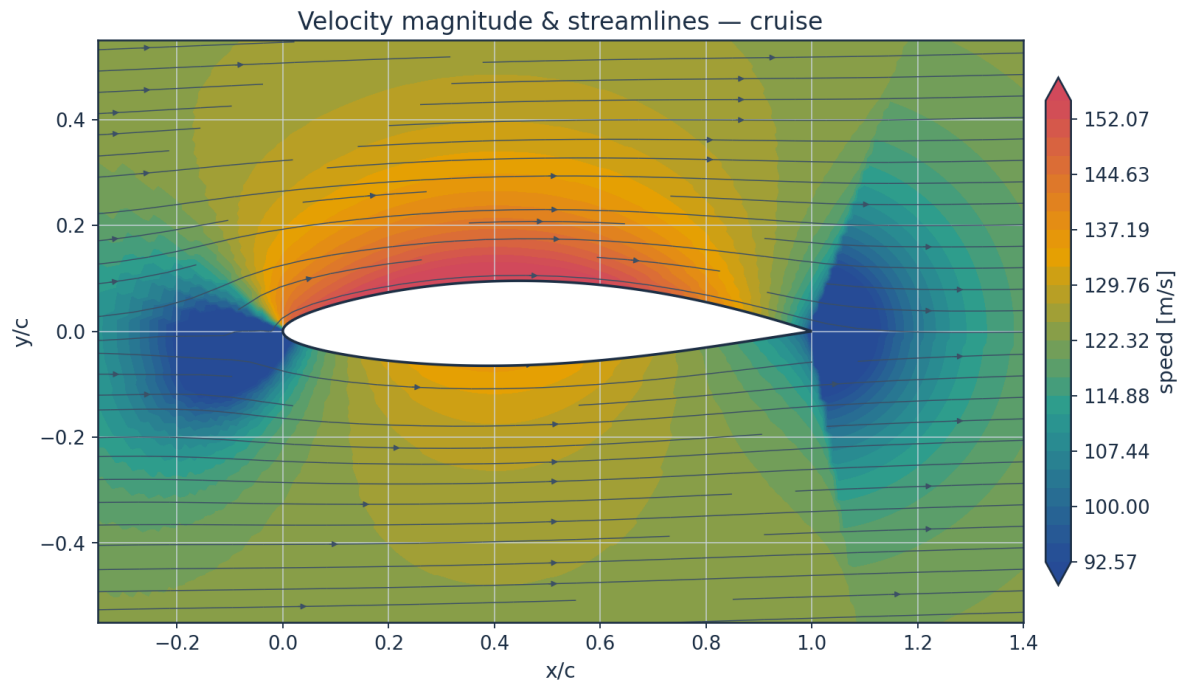


Fig. 25. Velocity magnitude and streamlines — cruise.

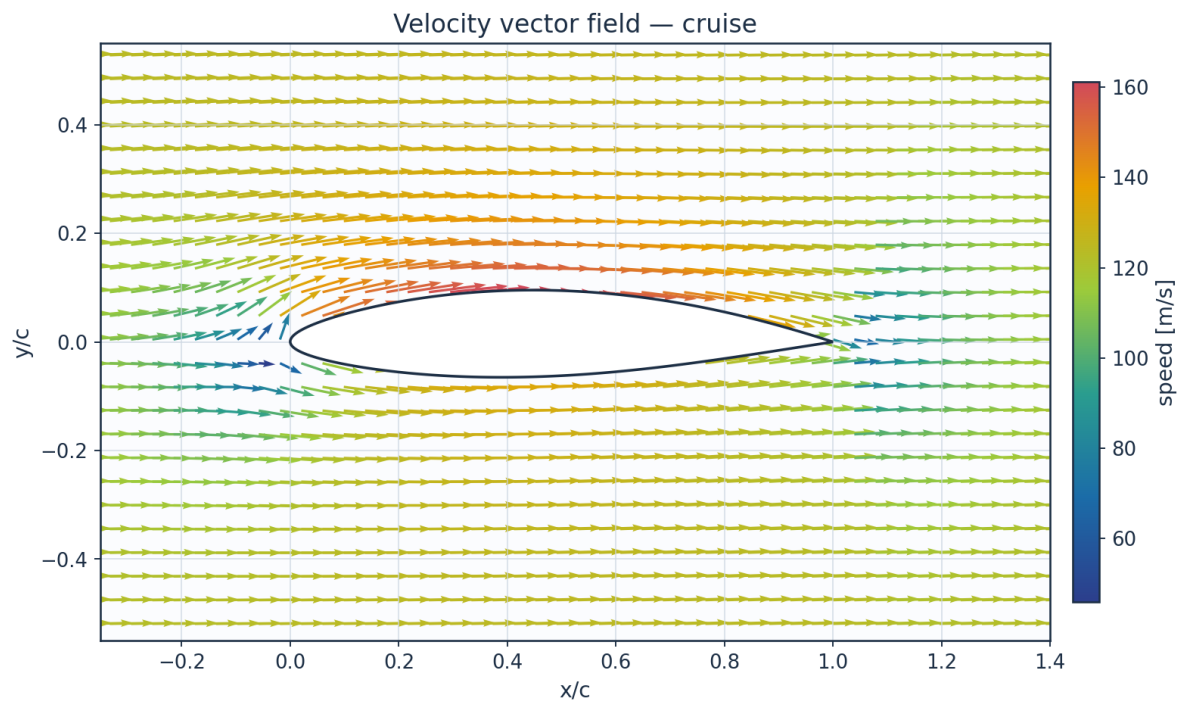


Fig. 26. Velocity vector field — cruise.

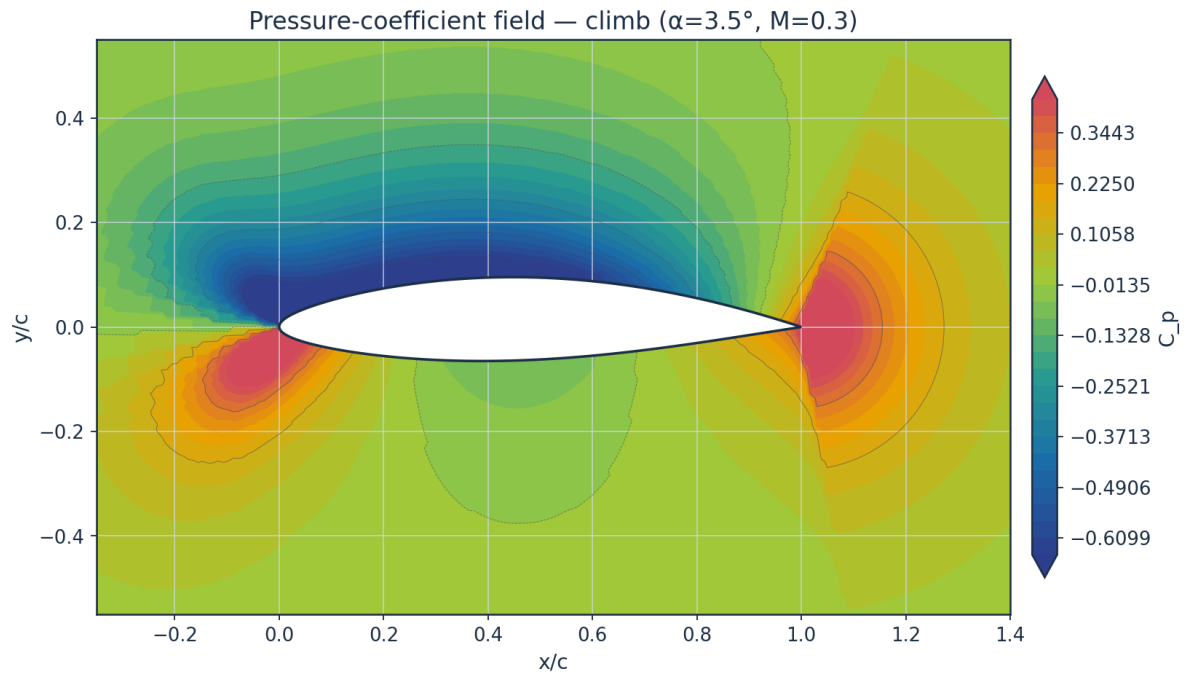


Fig. 27. Pressure-coefficient contour — climb.

10.2 Boundary-layer velocity and temperature profiles

Temperature profiles are obtained from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

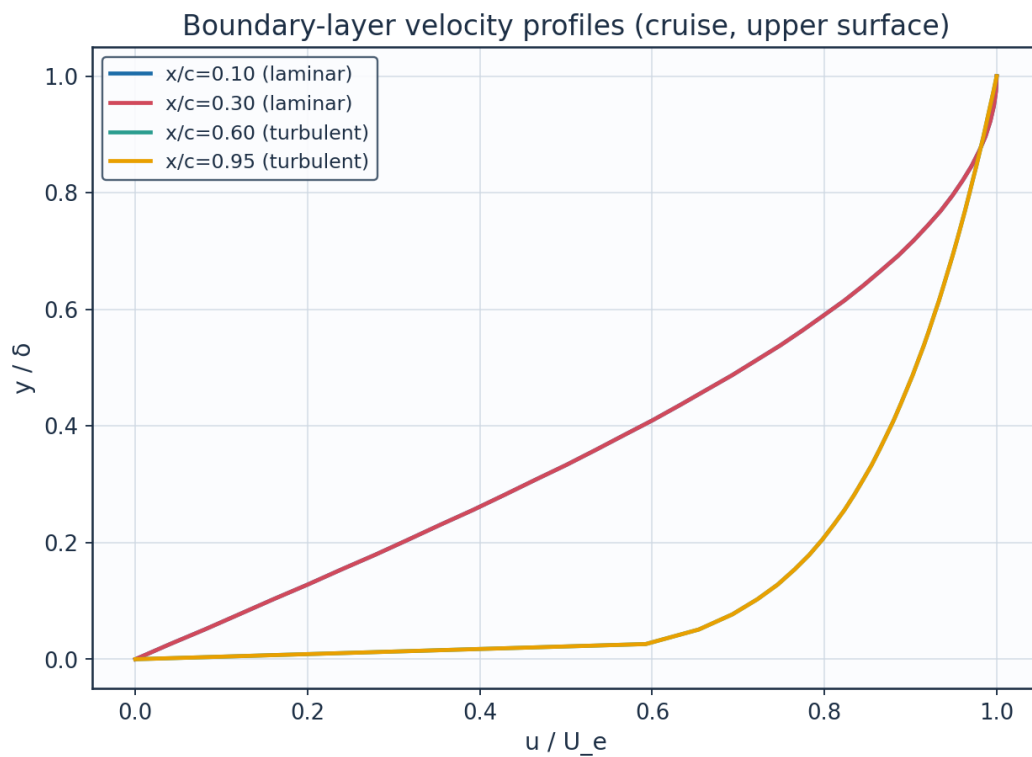


Fig. 28. Boundary-layer velocity profiles.

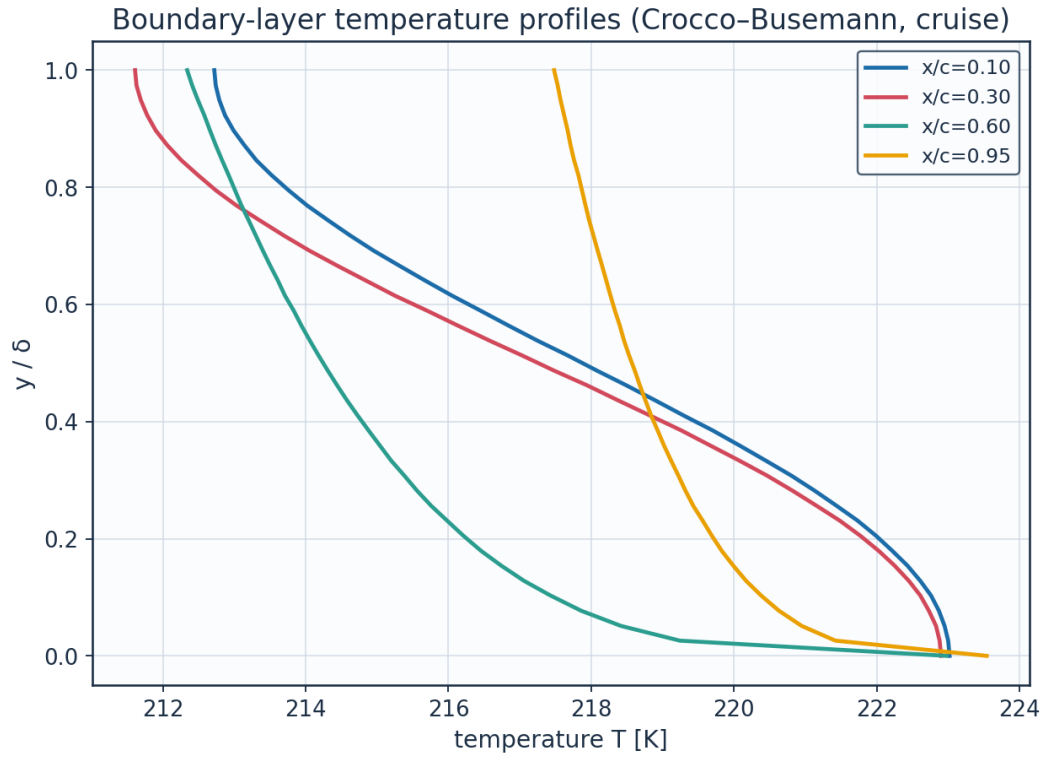


Fig. 29. Boundary-layer temperature profiles (K).

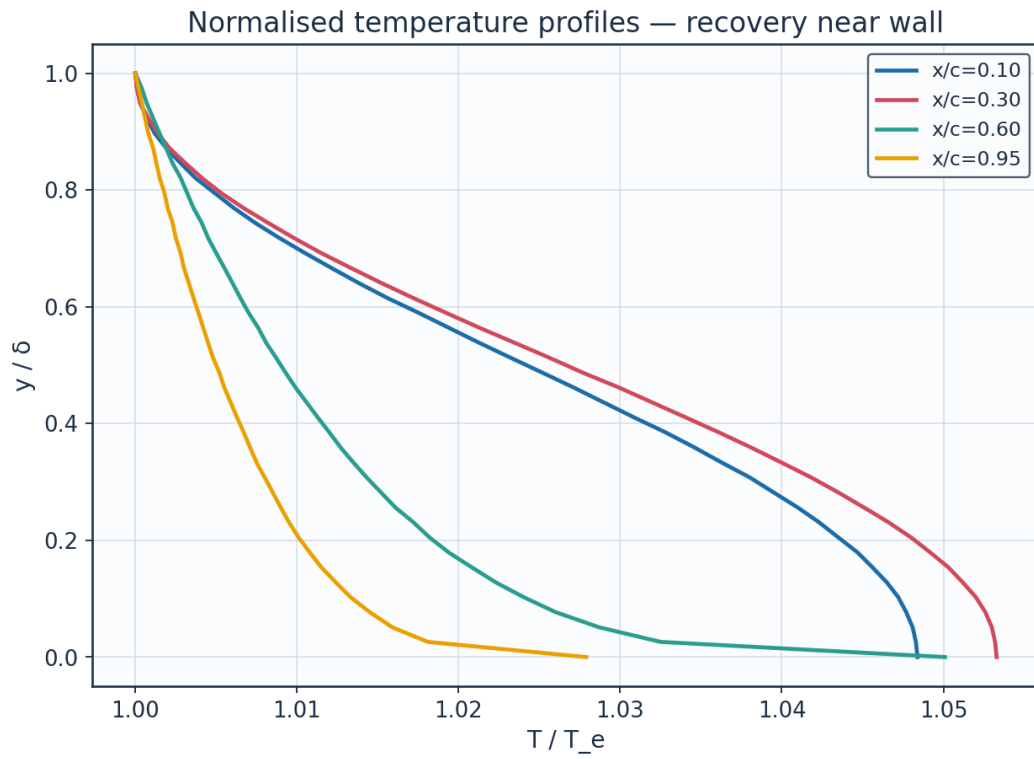


Fig. 30. Normalised temperature profiles T/T_e .

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K	Me_edge	state
x/c=0.10	0.1	0.0	0.0	0.0	1.0484	223.02	0.521	laminar
x/c=0.10	0.1	0.128	0.2496	0.2	1.0465	222.61	0.521	laminar
x/c=0.10	0.1	0.256	0.4991	0.392	1.041	221.44	0.521	laminar
x/c=0.10	0.1	0.385	0.7487	0.5681	1.0328	219.7	0.521	laminar
x/c=0.10	0.1	0.513	0.9982	0.7212	1.0232	217.67	0.521	laminar
x/c=0.10	0.1	0.641	1.2478	0.8452	1.0138	215.67	0.521	laminar
x/c=0.10	0.1	0.769	1.4973	0.935	1.0061	214.02	0.521	laminar
x/c=0.10	0.1	0.897	1.7469	0.9871	1.0012	212.99	0.521	laminar
x/c=0.30	0.3	0.0	0.0	0.0	1.0533	222.9	0.547	laminar
x/c=0.30	0.3	0.128	0.4093	0.2	1.0512	222.45	0.547	laminar
x/c=0.30	0.3	0.256	0.8186	0.392	1.0451	221.16	0.547	laminar
x/c=0.30	0.3	0.385	1.2279	0.5681	1.0361	219.26	0.547	laminar
x/c=0.30	0.3	0.513	1.6371	0.7212	1.0256	217.03	0.547	laminar
x/c=0.30	0.3	0.641	2.0464	0.8452	1.0152	214.84	0.547	laminar
x/c=0.30	0.3	0.769	2.4557	0.935	1.0067	213.03	0.547	laminar
x/c=0.30	0.3	0.897	2.865	0.9871	1.0014	211.9	0.547	laminar
x/c=0.60	0.6	0.0	0.0	0.0	1.0501	222.98	0.531	turbulent
x/c=0.60	0.6	0.128	1.1317	0.7457	1.0223	217.06	0.531	turbulent
x/c=0.60	0.6	0.256	2.2633	0.8233	1.0161	215.76	0.531	turbulent
x/c=0.60	0.6	0.385	3.395	0.8724	1.012	214.88	0.531	turbulent
x/c=0.60	0.6	0.513	4.5267	0.909	1.0087	214.18	0.531	turbulent
x/c=0.60	0.6	0.641	5.6583	0.9384	1.006	213.61	0.531	turbulent
x/c=0.60	0.6	0.769	6.79	0.9632	1.0036	213.1	0.531	turbulent
x/c=0.60	0.6	0.897	7.9217	0.9847	1.0015	212.66	0.531	turbulent
x/c=0.95	0.95	0.0	0.0	0.0	1.0279	223.54	0.396	turbulent
x/c=0.95	0.95	0.128	5.6506	0.7457	1.0124	220.17	0.396	turbulent
x/c=0.95	0.95	0.256	11.3012	0.8233	1.009	219.43	0.396	turbulent
x/c=0.95	0.95	0.385	16.9517	0.8724	1.0067	218.93	0.396	turbulent

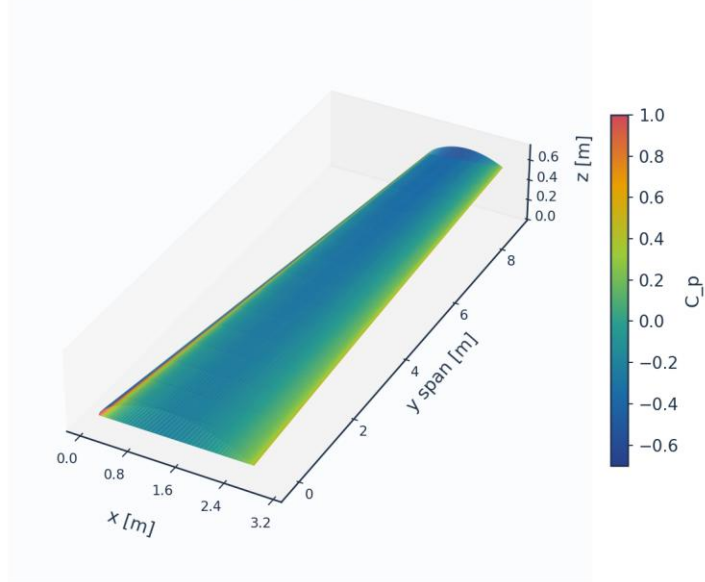
Table 19. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled).

(table sampled to 28 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

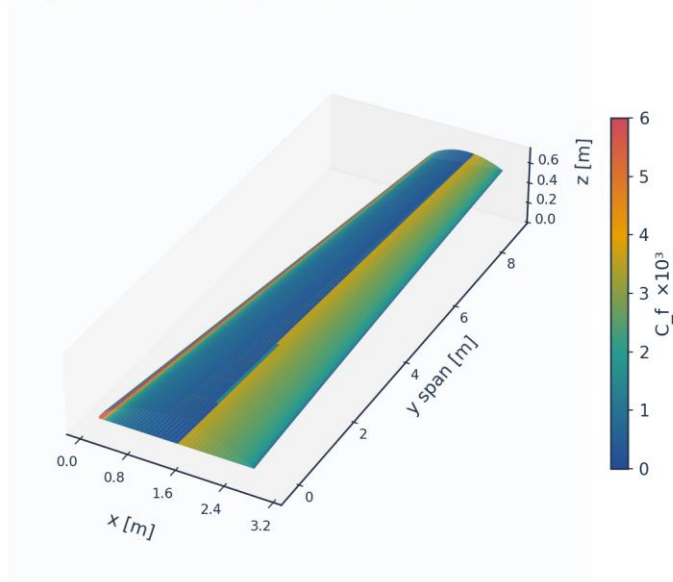
3D wing surface contour: C_p (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 31. 3-D wing surface contour — pressure C_p .

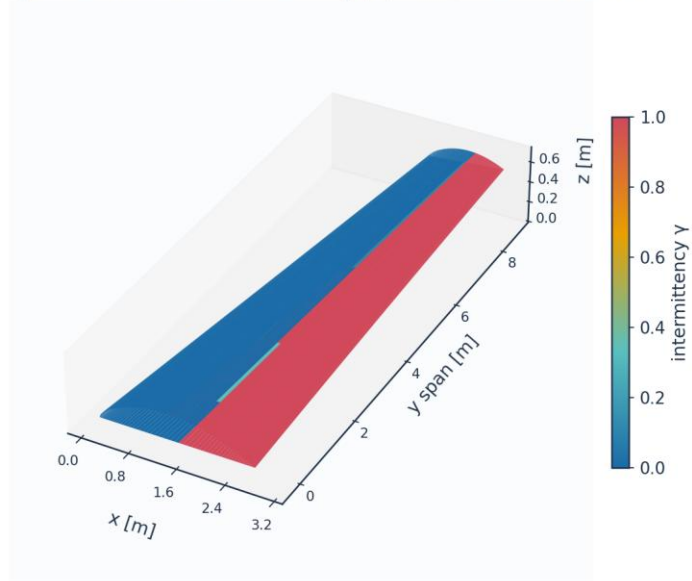
3D wing surface contour: $C_f \times 10^3$ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 32. 3-D wing surface contour — skin friction $C_f (\times 10^3)$.

3D wing surface contour: intermittency γ (cruise) — AETHER-NLF 25



Upper+lower surfaces, lofted UTSS-NLF16 sections; cruise FL360 M0.42

Fig. 33. 3-D wing surface contour — intermittency γ (transition front).

3D skin-friction vectors on upper surface (coloured by C_f)

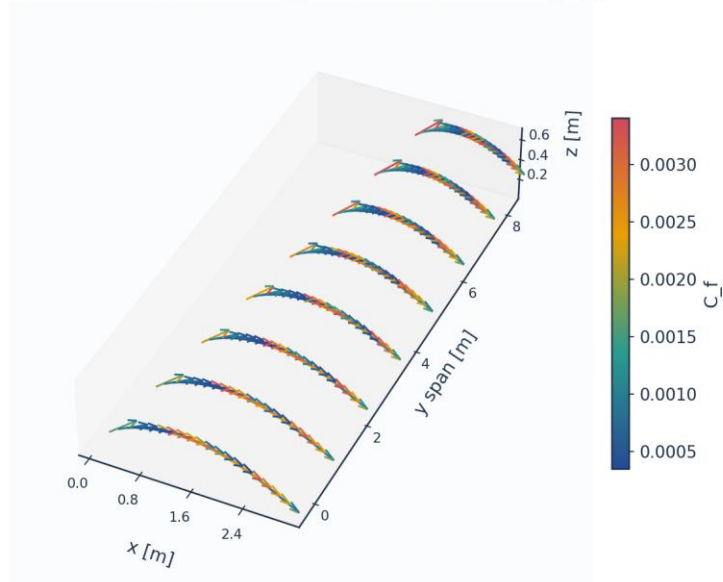


Fig. 34. 3-D skin-friction vectors on the upper surface.

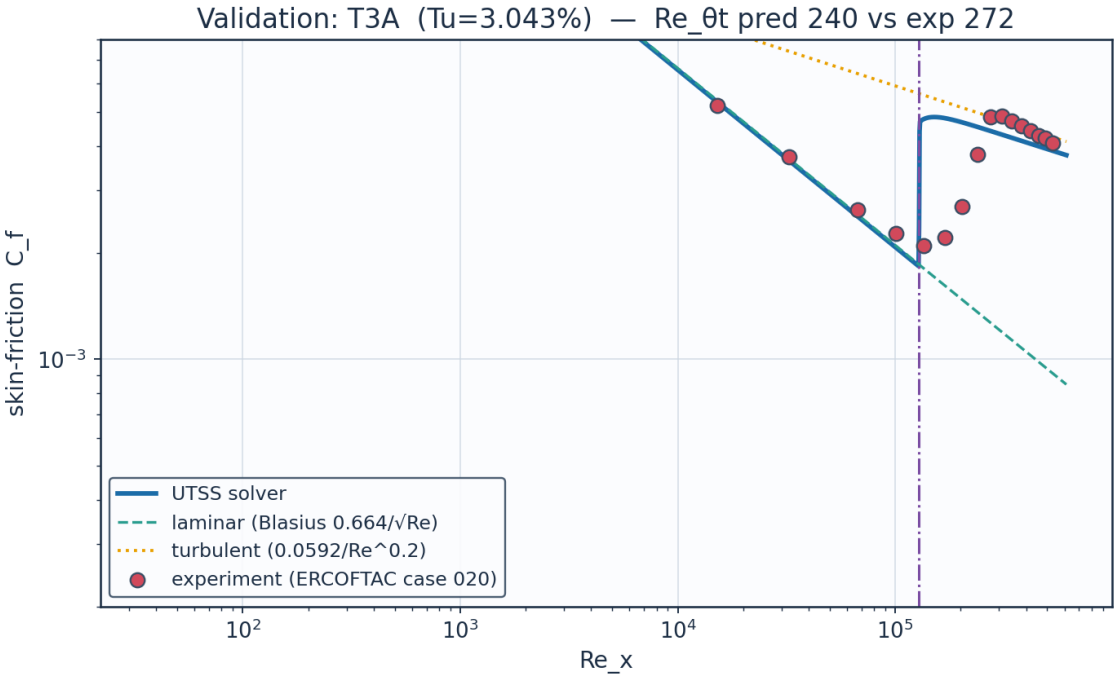
11. Validation and Calibration

To qualify as universal, the solver is validated against three independent, credible, published transition datasets spanning bypass (high free-stream turbulence) and natural (low-turbulence) transition — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number $Re_{\theta t}$ is the primary validation metric.

case	Tu_inlet_pct	L_turb_m	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_x_tr_exp	Re_x_tr_pred	mean_Cf_err_pct	Re_theta_meas_over_blasius	mechanism
ERC OFTAC T3A flat plate (bypass transition)	3.043	1.5299999999999998	272.3	240.0	-11.9	134800.0	128000.0	26.2	1.117	bypass
ERC OFTAC T3A-flat plate (low-Tu bypass)	0.874	0.98	818.8	636.3	-22.3	1443000.0	899800.0	247.6	1.027	bypass
ERC OFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	166.5	-8.2	591000.0	616800.0	10.4	1.123	bypass
ERC OFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	303.0	-20.5	178500.0	183400.0	54.2	1.359	separation
Schubaue & Skramstad flat	0.03		1100.0	1113.4	1.2	2800000.0	2755000.0			TS-natural

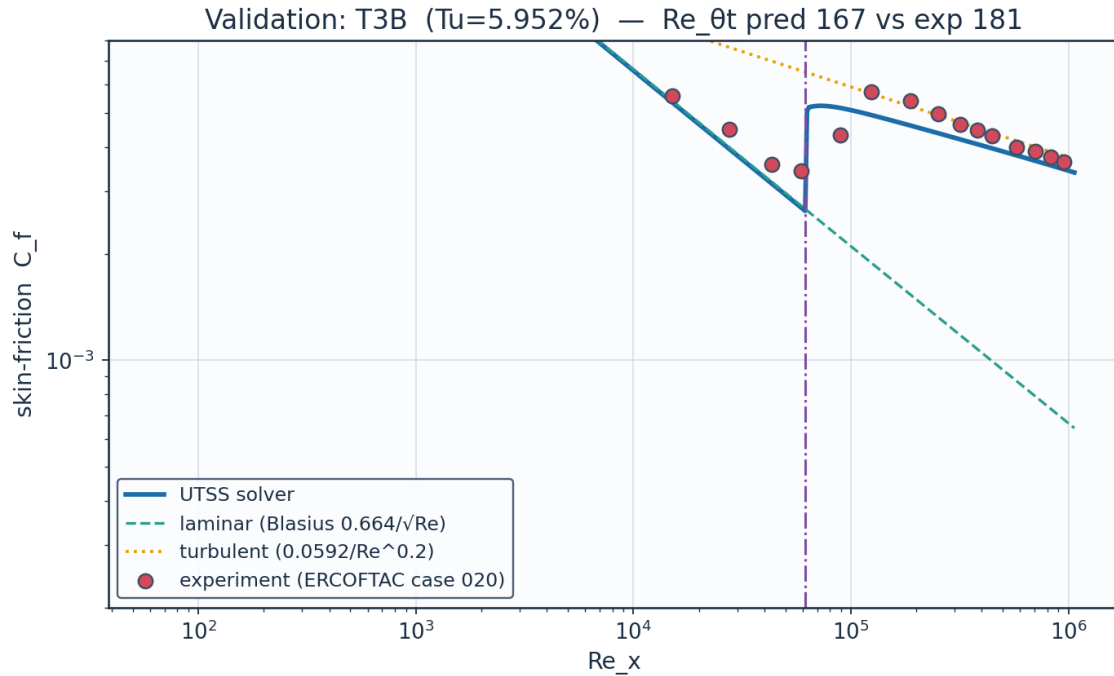
plate (natu ral trans ition)										
--	--	--	--	--	--	--	--	--	--	--

Table 20. Validation summary — predicted vs published $Re_{\theta t}$.



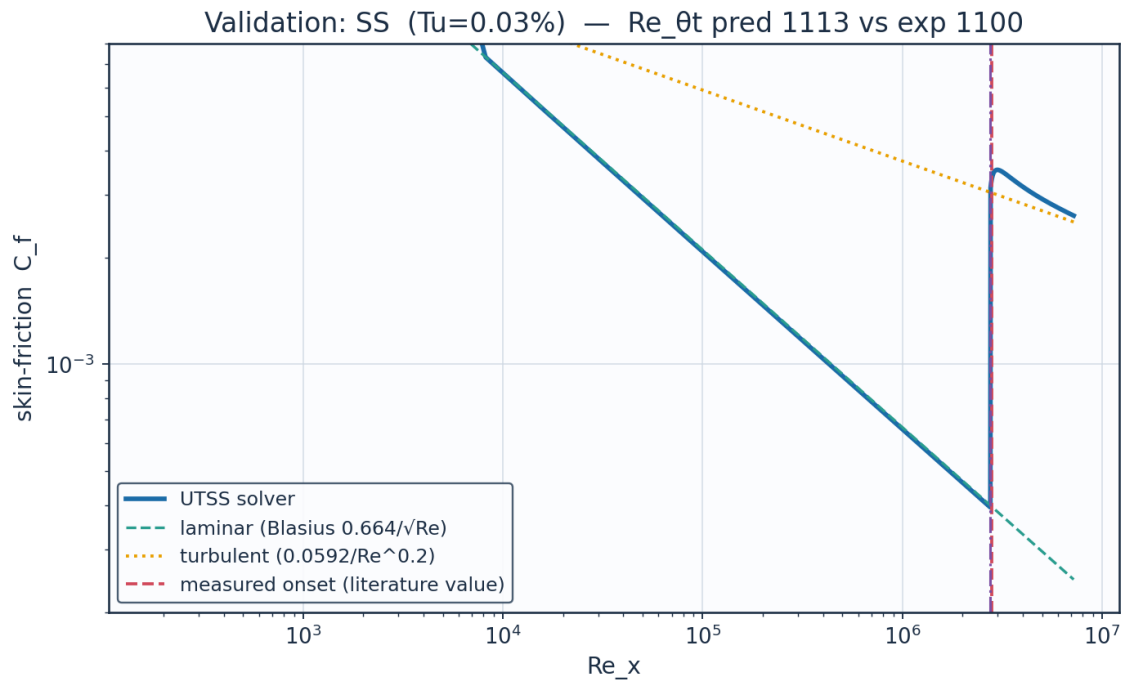
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 35. Validation — ERCOFTAC T3A flat plate (Tu=3.3 %).



Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009....

Fig. 36. Validation — ERCOFTAC T3B flat plate (Tu=6.0 %).



Source: Schubauer & Skramstad (1948), NACA Report 909....

Fig. 37. Validation — Schubauer & Skramstad natural transition.

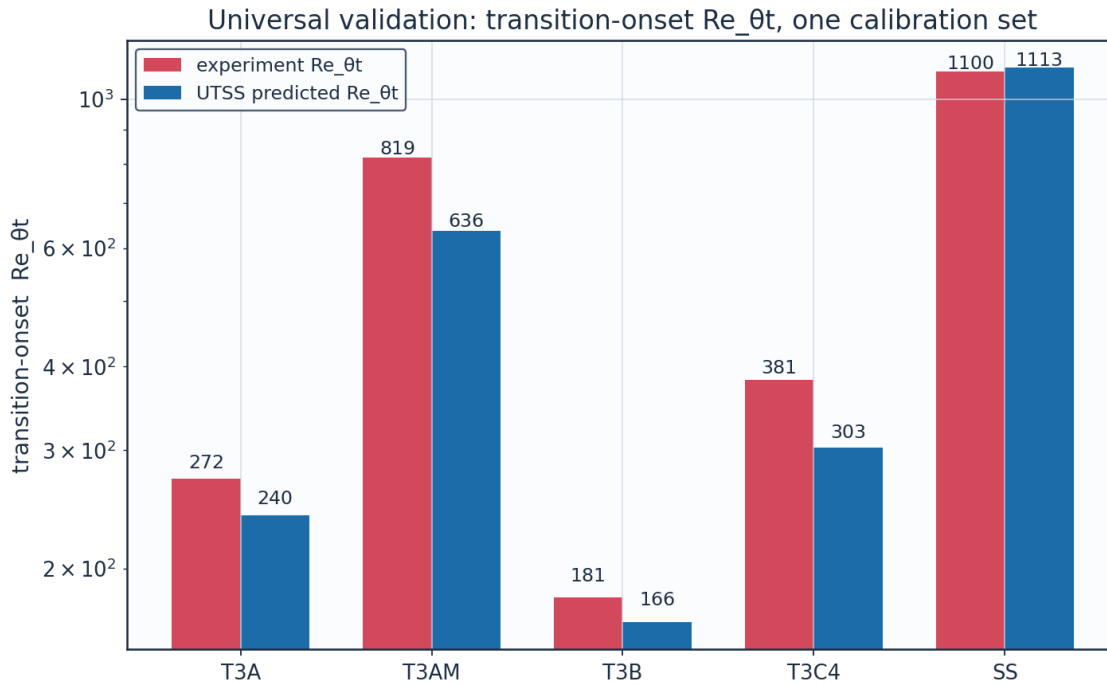


Fig. 38. Universal validation — $Re_{\theta t}$ across all cases.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted, and so returns 8.68 for any stream quieter than 0.08 %. The SAME constants reproduce every validation dataset — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

11.1 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.
Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B ($Tu=6\%$); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909.

Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwig H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and experimental results for a natural-laminar-flow aerofoil for general aviation applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 21. Validation, calibration and case-study sources.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by a minimum-effective-onset rule — reproducing all regimes with one calibration set.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at $Re_{\theta} = 200$ against the accepted 200.5.
- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_{crit} used elsewhere, so the length scales with the disturbance environment.
- Regime coverage with one constant set: transition-onset $Re_{\theta,t}$ predicted to -8.2 % on T3B, -11.9 % on T3A, -20.5 % on T3C4, -22.3 % on T3A- and +1.2 % on Schubauer-Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.040 chord,

and 38 of the 86 predictions fall inside the $\pm 0.025c$ bracket within which the experiment itself localises transition.

- A robust, panel-method-cost (< 1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry $\rightarrow$ mesh $\rightarrow$ setup $\rightarrow$ solution $\rightarrow$ post-processing $\rightarrow$ validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: ≈ 53 % laminar flow and ≈ 43 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans 0.03-6 % free-stream turbulence intensity. At cruise the wing achieves ≈ 52 % laminar flow and a ≈ 43 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 50 % larger than the first. And the method is an attached-flow formulation with an incidence envelope of about ± 8.5 deg on a real section; beyond that the laminar march separates near the leading edge and the result should not be relied on.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 104 (CSV: 43, PNG: 59, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV
01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG

01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_climb.npz	NPZ
04_solution/field_pressure_cruise.csv	CSV
04_solution/field_pressure_cruise.npz	NPZ
04_solution/integrated_forces.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG
05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG

05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/experiment_SS.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG
06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV

Table A1. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m ²
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Section design Cl	0.45	-

Fig. B1. geometry_definition.csv.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	321	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	6.51	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.099	mm (xc)
Max streamwise spacing	9.927	mm (xc)
LE clustering ratio	99.9	-
BL grid outer extent	7.04	mm
Friction velocity u_tau (TE)	4.800	m/s
Max cell aspect ratio	1524	-
Grid orthogonality (min)	88.5	deg
Mesh type	structured C-type (surface) + normal reconstruction	-

Fig. B2. mesh_metrics.csv.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-5	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor (turbulent)	0.89	-
Cruise dynamic viscosity	1.422e-5	Pa.s
Cruise density	0.36392	kg/m ³

Fig. B3. material_properties.csv.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility	none - incompressible panel solution
Laminar BL closure	Thwaites integral method
Transition model	UTSS unified 4-mechanism kernel
mechanisms	TS/natural(AGS@Tu=0.03%) bypass(AGS@Tu) separation crossflow(C1)
Intermittency closure	Narasimha universal (gamma)
Turbulent BL closure	Head entrainment + Ludwig-Tillmann Cf
Separation criterion	Thwaites $\lambda \leq -0.09$ / $H > 2.6$ turbulent
Drag integration	Squire-Young far-wake
Marching scheme	explicit, arc-length stepping
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	~1 (reconstruction grid)

Fig. B4. solver_settings.csv.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.5171	-
Section profile drag Cd	0.00496	-
Section Cd (counts)	49.6	counts
Section L/D	104.2	-
Wing lift (3D est, e=0.85)	43114.0	N
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6414000.0	-
Mach number	0.42	-

Fig. B5. integrated_forces.csv.